Predictive Maintenance of Power Transformers Using Machine Learning

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Abstract

Power transformers are critical components of electrical power systems, connecting generation, transmission, and distribution networks. Transformer failure carries severe operational and financial consequences. It can cost millions of pounds to replace just one large unit, and the loss of power during the replacement period affects both industrial and home users. Dissolved Gas Analysis has long been the industry standard for finding faults early. This method can be used when determining the signature gasses that are emitted in transformer oil at various fault types that are encountered. Even though DGA interpretation is widely used, traditional methods still rely heavily on fixed thresholds and ratio rules set by experts. These rules and thresholds don't work well in real-world multi-fault situations where things are complicated and change often.

The limitation of this project is mitigated by constructing and testing a machine learning pipeline on a publicly verified DGA dataset that consists of 2,321 labelled samples on seven different fault categories. The versions of classification algorithms, which are being utilized and tested under the same circumstances as the experiment, are: randomly chosen Forests, k-Nearest Neighbours, Decision Tree, Multi Layer Perceptron and Support Vector machine using radial basis function kernel. The imbalance between the classes based on fault types is dealt with the Synthetic Minority Over-sampling Technique that is only performed on the training partition to avoid the leakage of information.

Random Forest has performed best in all the assessments, with 85.96 percent accuracy in the held-out test set, macro F1-score of 0.784, and ROC-AUC of 0.973. This project not only classifies but also provides a fully operational pipeline: the probabilities of the classifier are turned into operational asset-level risk values, each of which is allocated to one of four levels of maintenance priorities and modelled into reactive and proactive maintenance cost estimates. The priority distribution can be adjusted to organisational workload constraints; threshold calibration makes this possible. With an assumption of the baseline economics, the structure estimates the net savings of about £707 million over the 2,321-asset portfolio as compared to a responsive maintenance baseline. The entire system is implemented as a Streamlit web application, which fulfils the requirement of the artefact of the BCS. The main value of the work is a traceable, explicit, and deployment-aware pipeline between machine learning fault classification and real operations maintenance decision-making, which, to the best of the available literature, has been discussed in parts only.

Keywords: Power Transformer, Dissolved Gas Analysis, Machine Learning, Random Forest, Predictive Maintenance, Fault Classification, SMOTE, Cost Modelling

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List of Abbreviations

Abbreviation	Full Term
ANN	Artificial Neural Network
BCS	British Computer Society
CO	Carbon Monoxide
CO_2	Carbon Dioxide
DGA	Dissolved Gas Analysis
F1	F1-Score (harmonic mean of precision and recall)
FCM	Fuzzy C-Means
GC	Gas Chromatography
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
k-NN	k-Nearest Neighbours
LSTM	Long Short-Term Memory
MLP	Multi-Layer Perceptron
ML	Machine Learning
OvR	One-vs-Rest
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
RBF	Radial Basis Function
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TDCG	Total Dissolved Combustible Gases
UI	User Interface

Chapter 1 : Introduction

1.1 Background and Motivation

The world electrical power system is reliant upon unprecedented numbers of transformers installed worldwide that are in constant operation at transmission and distribution systems. The assets cannot be easily substituted and the process of designing, manufacturing and commissioning a large power transformer is a time-consuming procedure and lead times between order and delivery can take several months and sometimes longer than one year. The cost of a unit of capital to provide a single unit of transmission level can go up to several million pounds and the economic implication of the unplanned outage of that waiting period; lost production, emergency stock supplies, fines by regulatory authorities and loss of reputation itself can overshadow the hardware cost.

To increase the economic pressure is the demographic reality of the installed fleet. In the developed power systems, the in-service age of transformers is much older than the original design of about 25 years and many transformers are well in their fourth or fifth decade of operation. The age increases the rate of degradation that causes the fault progression. Through thermal cycling, insulation materials continually enlarge and contract, moisture fluxes into the cellulose paper winding insulation with time, partial discharge activity destroys the dielectric integrity of oil and paper and mechanical stresses induced by fault currents undermine clamping and structural components. The net impact is that there is a significantly high likelihood of fault event from a cumulatively ageing fleet, as compared with a mid-life fleet.

The traditional reaction to this case has been regular checkups and routine change which is a costly method when used all over and inadequate when used selectively based solely on the age factor. Condition based maintenance is a better choice and is achieved through sustained or periodic monitoring of indicators that report the actual state of individual assets in a form where any maintenance can be planned to be necessary, but not when.

Dissolved Gas Analysis takes the centre stage of condition monitoring practise of transformers. The physical theory is simple: once the insulating oil or the cellulose paper in a transformer is put under some thermal or electrical stress out of the ordinary working limits, the decomposition responses produce characteristic gases which dissolve in the oil. The nature, location and extent of that fault mechanism is encoded in the type and relative proportions of such gases. DGA has a history in industrial applications spanning more than five decades, with the international standards which organise its interpretation, which are the IEC 60599:2022 and IEEE Std C57.104-2019, representing the accumulated engineering knowledge of that period.

Nevertheless, there are obvious and well-documented limitations of the conventional methods of interpretation. Comparable methods include the Duval Triangle, Rogers Ratio Method, Doernenburg Ratio Method and IEC ratio code, which involve comparison of measured gas ratios to prescribed ranges of threshold. All these thresholds were based on the population level data, and they do not modify to the specific design history, loading regulation, or ageing

peculiarities of transformers. In circumstances where a transformer exhibits several simultaneous fault types, as is common with aged equipment, the results of the ratio-based methods tend to give indeterminate or conflicting results. With mixed-fault conditions, these conventional methods are regularly reported to give classification accuracy of 60 to 80 percent.

Machine learning is a structurally unique solution. Instead of casting a hand-crafted rule structure onto the data, the supervised ML algorithms discover the relationship between levels of the gases and fault class directly by example. This property of it being a data-based model makes it possible to represent the non-linear relationships of the profiles of gases and the fault type too complicated and not able to be coded out in accordance with a set of established rules in ratios. The literature of the last ten years has shown multiple times that ML classifiers achieve better performance when compared to traditional tools on the benchmarks of DGA classification, with accuracy scores varying between the mid-80s with a basic implementation and above 96 percent with optimised ensemble tools and class balancing.

Yet there is still a gap despite this development. Most of the published research follows the approach of considering the problem as a classification exercise and leaves the research at the stage of coming up with a fault label. They fail to provide the next question, which is the question that concerns the maintenance engineer: given this fault classification, what should be done, when, and at what projected cost? This project seals that gap through construction of an entire pipeline from raw DGA measurements all the way to maintenance priority assignment and projection of financial impact, delivered as an interactive deployment-ready application.

Figure 1.1 : High-level system pipeline

1.2 Problem Statement

The present project is related to two issues. The former is the technical drawback of the current DGA interpretation: with standard approaches, in which gas ratios and fixed thresholds are used to define transformer faults, the classification of transformer faults is not reliable under real-world conditions, especially in situations where gas signatures caused by several interacting fault mechanisms are mixed. The second one is the operational gap in the research literature: although the research has shown that machine learning has boosted the accuracy of classification, the published work does not go further to relate such predictions with maintenance planning nor with economic analysis. The net effect is that utilities that can access DGA data nevertheless do not have a systematic, auditable, data-driven basis on which they can attempt to prioritise maintenance resources and project a financial payoff of a predictive

maintenance programme. In this project, there is the development of a system that covers both, followed by their validation and deployment.

1.3 Research Objectives

The particular goals to be followed in the given project are: to perform an organised review of the theory of DGA interpretation, traditional techniques, and machine learning approaches as introduced into the literature; to discover, certify, and prepare a publicly reachable DGA dataset of large enough scope to conduct statistically significant multi-class supervised learning; to implement five machine learning classifiers in a reproducible experimental framework, applying stratified splitting and SMOTE-based class balancing to the training partition; to evaluate all models on a held-out test set using accuracy, macro-averaged precision, recall, and F1-score, along with per-class breakdowns and ROC-AUC; to design and implement a risk scoring engine that converts probabilistic classifier outputs into calibrated asset-level risk scores; to construct a maintenance prioritisation and cost projection framework, including threshold calibration against a target workload distribution and sensitivity analysis across economic parameter combinations; and to deploy the complete system as a Streamlit web application satisfying BCS artefact requirements.

1.4 Scope and Delimitations

The project uses a single labelled DGA dataset of 2,321 samples sourced from IEEE DataPort, containing five dissolved gas features: H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2. Carbon monoxide and carbon dioxide — which carry information about cellulose insulation degradation — are not present in the dataset and are therefore outside the scope of the classification model. The study does not incorporate real-time sensor feeds, time-series gas trend modelling, or transformer-specific engineering parameters such as voltage class, age, or load history. Cost projections use parameterised industry-derived figures and are intended as scenario analysis rather than audited financial forecasts. The classification task covers seven fault categories as defined by the dataset labels.

1.5 Original Contributions

The project contributes four different things to the field. First, it gives an empirically validated comparison of five machine learning algorithms in seven-class DGA fault diagnosis on a very large dataset, fully reporting per-class performance, compared to most of the published literature. Second, it trains a diagnostics-to-maintenance pipeline that accepts inputs of probabilistic classifier results and generates risk scores, priority grades, suggested activities, response targets, and estimates of costs as outputs — a chain of reasoning that existing classification studies do not extend to. Third, it generates quantified sensitivity analysis over 25 combinations of economic assumptions by means of a threshold calibration mechanism enabling the priority distribution to be adjusted to reflect the ability of an organisation to cope with maintenance workload. Fourth, it delivers the entire system as a user-facing web-based application by which maintenance engineers with no programming skills upload DGA data and get ranked maintenance recommendations in a format that can be used during operations.

1.6 Report Structure

Chapter 2 reviews the relevant literature, covering DGA principles, conventional interpretation methods, machine learning approaches, and the gap this project addresses. Chapter 3 describes the methodology: data acquisition, preprocessing, algorithm configurations, evaluation metrics, and cost modelling design. Chapter 4 documents the system architecture and artefact design. Chapter 5 presents experimental results and discusses their interpretation. Chapter 6 concludes with the statement of what has been done, the thoughts of what could be done and future work.

Chapter 2 : Literature Review

2.1 Introduction

The chapter creates the intellectual groundwork of the project. It begins with the engineering case of the monitoring conditions of the transformer failure and condition monitoring capabilities in managing assets. It goes on to look at the physical concepts of DGA and interpretation criteria under which it is used. The following part of the study provides a survey of machine learning literature in the field of transformer fault diagnostics, presents the principal families of algorithms, benchmark findings, and methodological issues. An ultimate synthesis section identifies the gap that is driving the contributions of this project.

2.2 Transformer Failure: Engineering Context and Economic Consequences

Power transformers are the most complicated and the expensive equipment within the electrical power system. They work under constant thermal and electrical loads throughout service life, often prolonged way beyond design values. The main driver of the failure is the degradation of the insulation that occurs because of the cumulative effects of thermal ageing, moisture ingress, partial discharge activity, and mechanical stress from fault currents and vibration.

The impact of transformer breakdown is experienced at different levels. The failure of a transmission transformer at the network level can be disastrous in terms of impacting a wide area of supply and may need costly emergency bypassing. At the asset level, replacement cost and lead time for a large unit put a financial strain that can last months. Unplanned downtime losses in an industrial context can be built up very quickly, especially in continuous-process industries where the failure of a transformer causes instant production to be halted.

Condition monitoring gives the ability to intervene before failure occurs. There is a variety of diagnostic methods practised, such as the analysis of oil quality, dissolved gas analysis, the measurement of furanic compounds, power factor measurements, partial discharge measurements, thermal imaging and frequency response analysis. All these methods observe a distinct element of transformer well-being, and their results are usually combined into a compound Health Index to provide a summary of asset state. The utilities have a challenge to find a systematic method through which they extract actionable maintenance decisions out of this multi-parameter diagnostic picture in a cost-rational manner.

(Islam et al., 2017; Yahaya et al., 2018a)

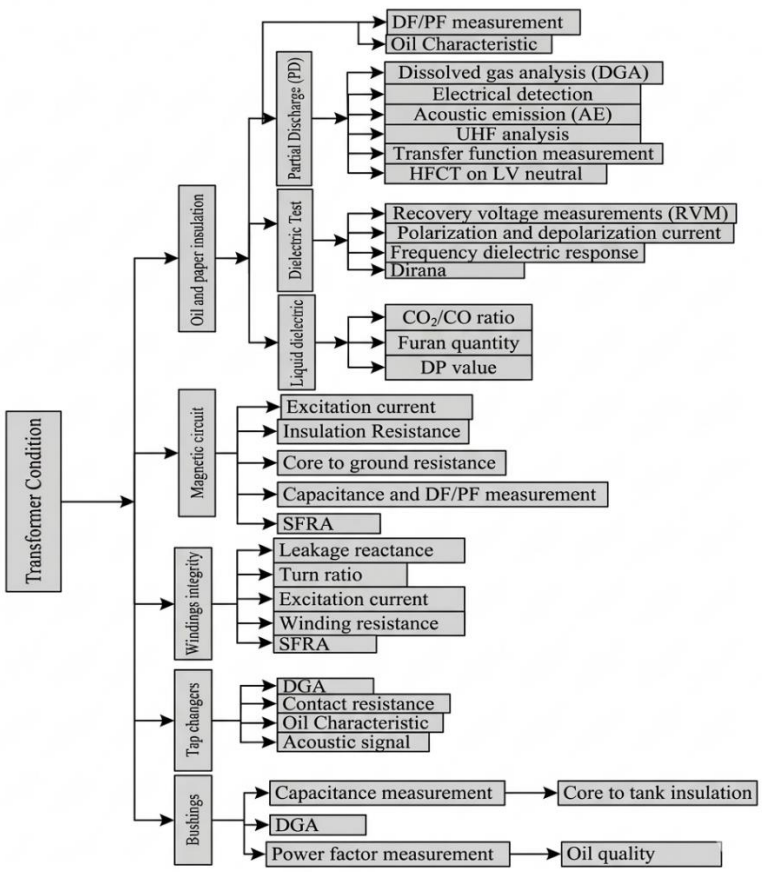

Figure 2.1 : Overview of power transformer condition monitoring and diagnostic techniques (Source: Islam, Lee and Hettiwatte, 2017)

2.3 Dissolved Gas Analysis: Physical Basis

Production of gases in transformer oil is a process that has a well-developed chemistry. Even when the mineral oil and cellulose paper insulation are operating within the normal operating conditions, the thermal decomposition proceeds slowly in low quantities, producing hydrogen, methane, ethane, ethylene and acetylene from oil decomposition, and carbon monoxide and carbon dioxide from paper degradation at normal operating conditions. These background quantities are regarded as normal and are factored in the diagnostic standards by using baseline threshold values.

The decomposition process increases in rate and the composition of the gases changes in characteristic ways when a fault develops, either thermally, electrically or a mixture of the two. The thermal faults can yield both methane and ethylene, as the proportion of ethylene to ethane rises with the elevation of the hotspot temperature. Partial discharge is an electrical fault that yields mainly hydrogen; while arcing events are characterised by the production of acetylene. The physical ground on which all DGA interpretation methods are based is the correlation between gas composition and fault type.

Transformer oil extraction is either carried out by standard laboratory analysis, in which an oil sample is taken and carried to the laboratory to be processed by chromatography, or via online

monitoring that continuously returns a low flow of oil through a gas-sensing device. Each method has its sources of uncertainty: the laboratory analysis creates the risk of gas escape and air intrusion when handling and transporting the samples, and online monitors may become biased after several years as the sealing mechanisms age. It has been demonstrated that results for one and the same transformer online and in the lab sometimes vary by up to 30 percent, and this uncertainty of measurement has consequences for the reliability of any subsequent method of interpretation. The normal gas chromatography analysis contains the concentrations of seven major gases, measured in parts per million: hydrogen, methane, ethane, ethylene, acetylene, carbon monoxide, and carbon dioxide. The individual concentrations and combinations of these gases create the feature set on which DGA interpretation activity is performed, whether by conventional methods or by machine learning.

(Bakar, Abu-Siada and Islam, 2014; Wei, Long and Yan, 2017; Rangel Bessa et al., 2023)

Table 2.1 : Key Dissolved Gases and Associated Fault Mechanisms

Gas	Chemical Formula	Primary Association
Hydrogen	H_2	Partial discharge, low-energy electrical activity
Methane	CH_4	Low-temperature thermal degradation below 300°C
Ethane	C_2H_6	Moderate thermal faults in the 300–700°C range
Ethylene	C_2H_4	High-temperature thermal faults above 700°C
Acetylene	C_2H_2	Arcing and high-energy electrical discharge
Carbon Monoxide	CO	Paper insulation (cellulose) degradation
Carbon Dioxide	CO_2	Oxidative cellulose breakdown

Source: Adapted from IEC 60599:2022 and IEEE Std C57.104-2019

2.4 Conventional DGA Interpretation Methods and Their Limitations

2.4.1 International Standards: IEC 60599:2022 and IEEE Std C57.104-2019

There are two main international standards that regulate DGA interpretation: IEC 60599:2022 and IEEE Std C57.104-2019. Both establish concentration limits of key gases and give ratio-based techniques of classifying faults. The IEC standard differentiates between five major types of faults that can be identified on a visual inspection following decommissioning and include: partial discharge, low-energy discharge, high-energy discharge, thermal faults below 700 degrees Celsius, and thermal faults above 700 degrees Celsius. IEEE Std C57.104-2019 uses a flow chart to understand the interpretation of the DGA results, defines concentration thresholds at the 90th and 95th percentile of the population, and includes guidance on gas rate-of-change analysis using multi-point linear regression.

The inherent weakness of these standards is that their thresholds and ratio limits are population-wide measures based on a wide cross-section of types of transformers, ages, and operating histories. When applied to an individual transformer with a specific design, loading profile and maintenance history, they will not be able to tell when a particular unit is showing normal variation, or when an early fault is in the early stages of development. This fundamental limitation is inherent to the standards-based approach and is not resolved by refinements to the ratio definitions.

(IEEE, 2019; International Electrotechnical Commission, 2022)

2.4.2 The Duval Triangle and Pentagon Methods

The Duval Triangle is a plot of the relative percentages of methane, ethylene, and acetylene as a single point in a triangular coordinate space subdivided into fault zones. The triangle is divided into seven labelled areas which represent the fault categories employed by the IEC standard. It is graphical in nature and hence intuitively accessible and has been widely used in conjunction with ratio-based methods. The Duval Pentagon generalises this to five gas dimensions by adding ethane and hydrogen as additional axes, enabling finer discrimination between subtypes of thermal faults.

There are limitations on the usefulness of these graphical techniques in practise. They merely take a subset of the available gas data and hence are not able to obtain the complete informational content of a DGA measurement. In mixed-fault conditions, where a transformer is subjected to both thermal and electrical stress, the plotted point can be in an ambiguous zone or at a zone boundary, generating uncertain diagnoses. Experiments with real-world fault data have verified that, in certain multi-class scenarios, the Duval Triangle and Pentagon reach a classification accuracy of only 30–35 percent, which is far short of what is required for reliable predictive maintenance.

(Duval and DePablo, 2001; Prasojo et al., 2023; Dladla and Thango, 2025)

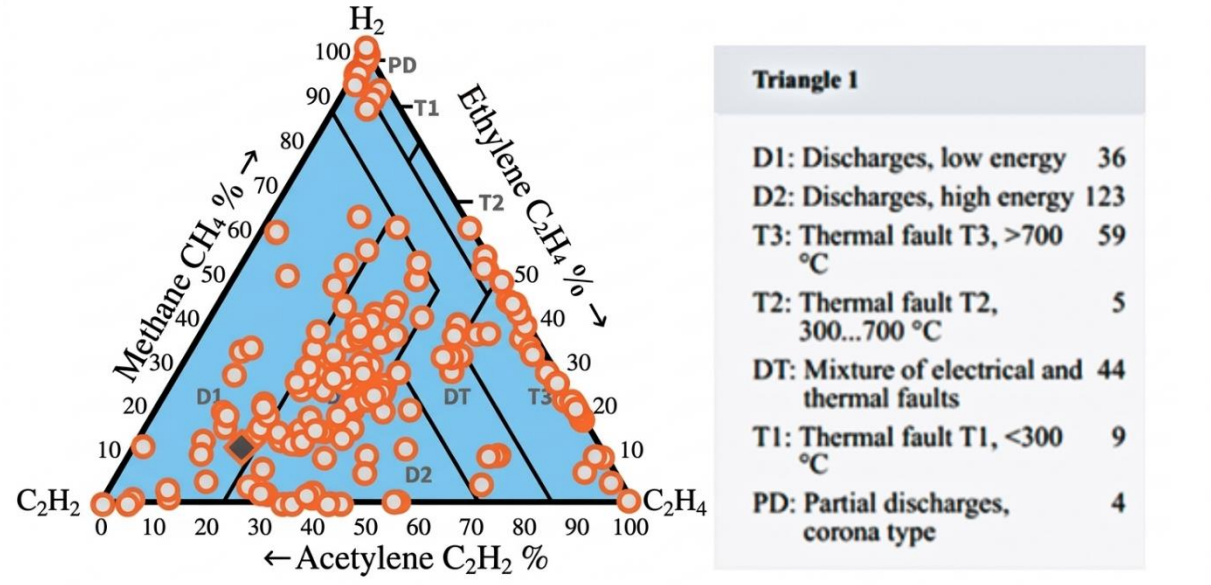

Figure 2.2 : The Duval Triangle (Source: Duval and DePablo, 2001)

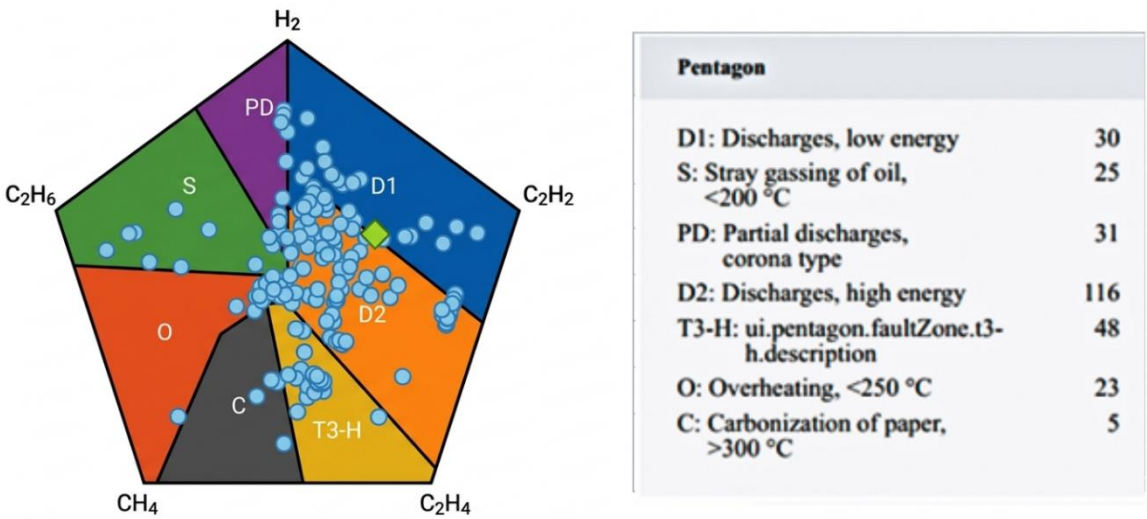

Figure 2.3 : The Duval Pentagon (Source: Prasojo et al., 2023)

2.4.3 Ratio-Based Methods: Rogers and Doernenburg

Rogers Ratio process involves the use of three ratios — acetylene to ethylene, methane to hydrogen, and ethylene to ethane — each of which is mapped to a coded value that combines to indicate a fault category. The Doernenburg method applies four gas ratios but only when at least one of the gas concentrations is found to be above a given minimum limit. The two methods have a common structural vulnerability in that they both work with static lookup tables, which were derived from historical fault data and cannot adjust to new patterns or update as more labelled examples become available.

The studies that explicitly tested these procedures under measurement uncertainty conditions established that certain boundary zones in the Duval Triangle become unreliable under a ±10 percent measurement error typical of laboratory analysis, and that ratio-based procedures have missing interpretation codes that prevent fault identification in a non-trivial fraction of cases. Attempts to make the methods more accurate by altering ratio code limits yielded accuracy improvements but did not eliminate the fundamental adaptability limitation.

(Mirowski and Lecun, 2012; Taha, Ghoneim and Zaini, 2015; Wei, Long and Yan, 2017)

2.4.4 Software Tools for Conventional DGA

Several software suites have been written to implement traditional DGA interpretation, among these commercial systems like Serveron, Inside View and GE Perception Fleet, and the research-oriented system DGALab. Academically, DGALab is interesting in that it has a single framework that includes several different methods of DGA, such as the Duval Triangle, the Rogers four-ratio, IEC 60599, as well as improved forms of each, and has an identical output specification that allows reasonable comparative accuracy analysis. Nonetheless, even these software programmes are confined solely to rule-based interpretations and lack machine learning elements that would learn from accumulated diagnostic data or extend naturally to maintenance cost projection.

(Ibrahim, Ghoneim and Taha, 2018)

Table 2.2 : Summary of Conventional DGA Interpretation Methods

Method	Gas Inputs	Approach	Primary Weakness
Duval Triangle	CH_4, C_2H_4, C_2H_2	Graphical zone mapping	Poor performance on mixed faults
Duval Pentagon	H_2, CH_4, C_2H_6, C_2H_4, C_2H_2	Extended graphical	Static zone boundaries
Rogers Ratio	Three gas ratios	Coded lookup table	Missing codes, no adaptability
Doernenburg	Four gas ratios	Threshold-triggered lookup	Requires minimum gas levels
IEC 60599:2022	Multiple gases	Population thresholds	Not transformer-specific
IEEE C57.104-2019	Multiple gases	Percentile thresholds + rates	Population-level averaging

Source: Compiled from Duval and DePablo (2001), IEC (2022), IEEE (2019), Rangel Bessa et al. (2023)

2.5 Machine Learning Approaches to DGA Fault Diagnosis

2.5.1 The Shift to Data-Driven Classification

The case to abandon or complement rule-based interpretation of DGA with machine learning is based on a straightforward observation: the dependence between gas levels and fault type is complex, non-linear and immensely diverse over transformer models and operating histories. Having such a model learn this mapping based on labelled data, instead of expert knowledge, is better placed to capture that complexity.

An extensive early analysis which compared fifteen conventional statistical learning algorithms at the DGA classification tasks revealed that nonlinear models, namely support vector machines using Gaussian kernels, neural networks and local linear regression, performed better than linear classifiers in this problem. That review also found a critical preprocessing step: applying a logarithmic transformation to gas concentration data before training, as the raw distributions of concentration are severely right skewed. This preprocessing insight — that the feature space contains structural properties that require transformation before distance-based or gradient-based learning — remains relevant for the algorithms implemented in this project.

(Mirowski and Lecun, 2012)

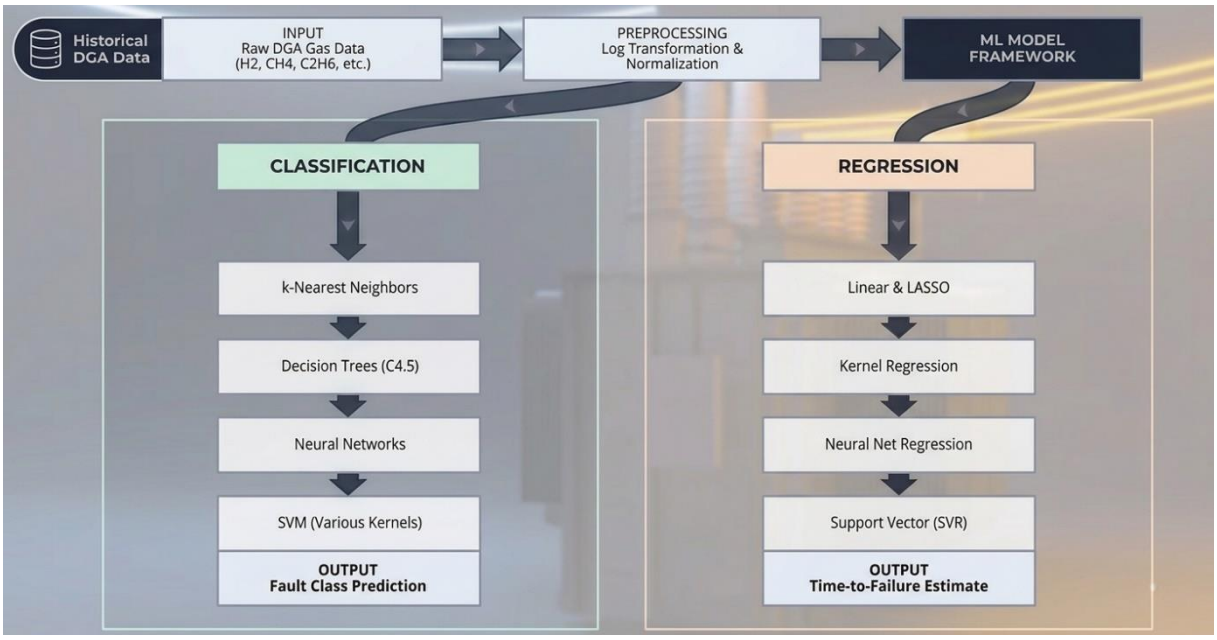

Figure 2.4 : Taxonomy of statistical machine learning methods applied to DGA-based transformer fault diagnosis (Source: Mirowski and LeCun, 2012)

2.5.2 Support Vector Machines

SVM builds a decision boundary in a kernel-transformed feature space that maximises the classification margin. Radial basis functional kernel is particularly ideal to DGA classification

because it can help to depict the non-linear complex classifications of the boundaries that manifest when gas signature of different types of fault intersect on a concentration space. Initial use of SVM on DGA showed that it worked better than traditional ratio-based methods, particularly at the detection of partial discharge events where hydrogen is the major gas and a distinct separating feature.

Subsequent research extended the SVM method with optimisation strategies. Genetic algorithm-based hyperparameter optimisation on the IEC TC10 dataset gave about 87 percent accuracy; particle swarm optimisation combined with Duval Pentagon features reached 88 percent; and a Krill Herd-optimised SVM achieved 85.71 percent in a cross-validated evaluation. This body of work is generally consistent in the message that performance of the SVM is very sensitive to hyperparameter control and the specific dataset used, and these facts have direct consequences for interpretation of the baseline SVM results in this project.

(Bacha, Souahlia and Gossa, 2012; Dladla and Thango, 2025)

Figure 2.5 : SVM-based DGA fault classification architecture (Source: Bacha, Souahlia and Gossa, 2012)

2.5.3 Random Forest

Random Forest is an ensemble classifier that builds a large collection of decision trees, each trained on a bootstrap sample of the data with a random subset of features considered at each split. The majority vote across all trees produces the final class prediction. The approach is ideally suited to DGA classification due to a number of factors: it is resistant to the scale of input features, thus it does not need standardisation; it gives feature importance scores that allow post-hoc interpretability of which gases are most diagnostic; and it can effectively deal with class imbalance through sample weighting options.

The best classification performance reached by a conventional machine learning method on DGA data was achieved with the merging of Random Forest and SMOTE preprocessing, which reached 96.2 percent on one of the fault classification schemes, and 96.5 percent on another. This experiment revealed that Random Forest with SMOTE outperformed six other ML algorithms in both accuracy and consistency across repeated evaluations under uncertainty. Further studies that incorporated Random Forest as the integration layer in a multi-method DGA interpretation framework — combining outputs from five conventional techniques including

Doernenburg, Rogers, IEC, Duval Triangle, and Duval Pentagon — confirmed that RF-based integration produced greater accuracy and consistency than any individual conventional method alone.

(Prasojo et al., 2023; Suwarno et al., 2024)

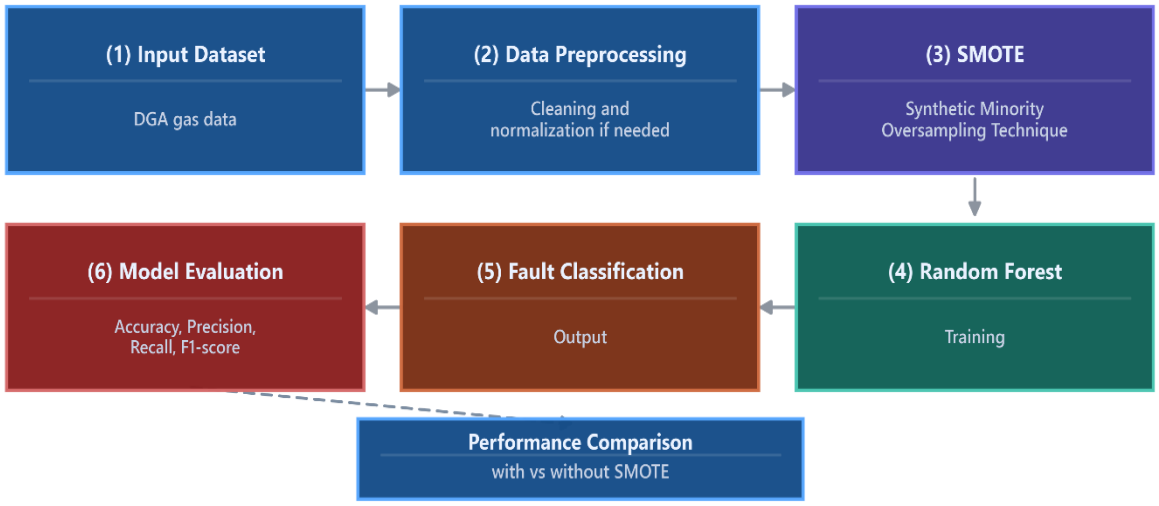

Figure 2.6 : : Random Forest with SMOTE methodology flowchart (Source: Prasojo et al., 2023)

2.5.4 k-Nearest Neighbours

k-Nearest Neighbours is an instance-based algorithm that classifies a new sample by looking up the k most similar training examples in feature space and assigning the majority class label among those neighbours. Its appeal in DGA classification lies in its simplicity and the intuitive correspondence between the algorithm's distance-based logic and the physical intuition that similar gas profiles should correspond to similar fault types.

An optimised k-NN realised using a Chi-Squared based feature selection and random search, with a hyperparameter exhibited a review accuracy of 92.478 percent the highest reported in the recent literature by traditional ML methods. The evaluation of the same study established that properly optimised k-NN substantially outperforms the Duval Triangle and Pentagon, which had only 35.757 percent and 30.818 percent accuracy respectively on the same dataset. k-NN is also prone to class imbalance since the local neighbourhood of a minority-class sample is often dominated by majority-class samples, and therefore SMOTE preprocessing before nearest-neighbour learning is particularly important for this algorithm.

(Dladla and Thango, 2025)

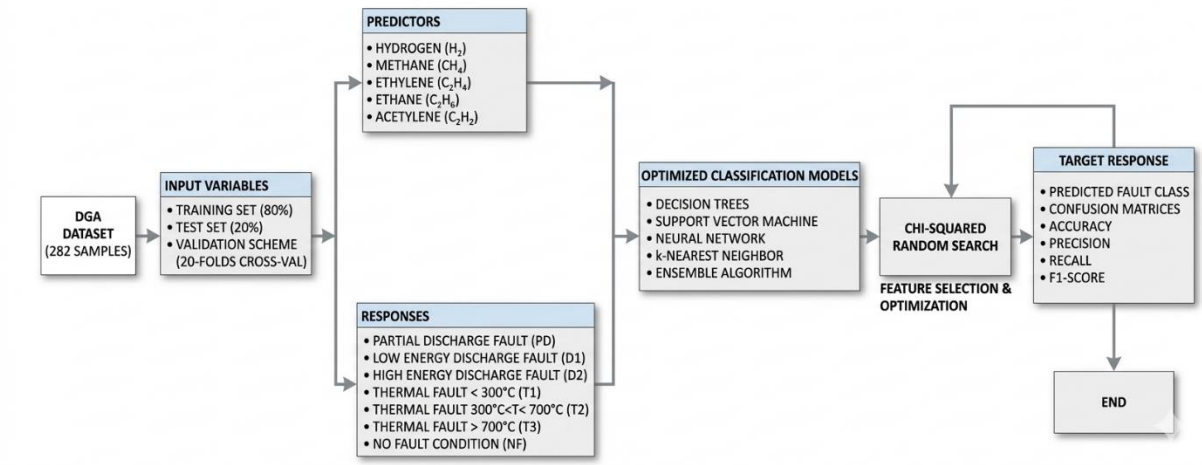

Figure 2.7 : Optimised machine learning pipeline for DGA fault (Source: Dladla and Thango, 2025)

2.5.5 Artificial Neural Networks and Deep Learning

Transformer fault diagnosis has a lengthy history of use of multi-layer perceptron networks. Early applications realised shallow architectures as extensions of expert systems, and they were mixed in performance. More recent optimised MLP implementations have accuracies of between 86 to 90 percent on standard DGA datasets. Depending on the amount of the layers, the number of neurons per layer, the activation mechanism and the regularisation policy, which influence generalisation, and on more advanced and larger data the comparative ability of the more elementary classifiers such as the Decision Tree are larger.

The highest attempt of applying neural techniques to the DGA classification is a Long Short-Term Memory architecture with a new meta-heuristic optimisation algorithm based on Gravitational Search and Dipper Throated Optimisation, achieving diagnostic accuracy of 98.26 percent. The GSDTO algorithm is applied to feature selection as well as optimisation of LSTM classification parameters, and ANOVA and Wilcoxon tests prove that the result is stable in respect to repeated assessments. The practical consideration for deployment contexts is that deep learning techniques of such sophistication need significantly more computational infrastructure and technical skills to run than ensemble or instance-based classifiers, which affects their suitability for end-user-facing applications.

(El-kenawy et al., 2022; Dladla and Thango, 2025; Saravanan, D and Vengateson, 2025)

2.5.6 Fuzzy Logic and Clustering-Based Approaches

Fuzzy logic offers a degree of membership in multiple fault categories, which is conceptually well-matched to the ambiguous, overlapping gas signatures characteristic of many real transformer faults. Fuzzy logic typically is used with traditional interpretation strategies in applying fuzzy logic to DGA — for example, using fuzzy membership functions applied to key gas, IEC ratio, and Duval Triangle outputs — and the subsequent fuzzy inference to generate a percentage failure index and a fault type determination. These systems have been tested on

datasets of several hundred transformer cases and have the benefit of providing graduated severity scores rather than discrete fault labels.

Clustering methods do not require labelled training data. The Fuzzy C-Means method divides DGA samples into clusters with each sample having a degree of membership in each cluster. The structural limitation of FCM for DGA data is that the standard membership function based on inverse Euclidean distance is non-monotonic with respect to sample distance, resulting in poor cluster separation for gas profiles that lie between fault categories. One of the suggested remedies — substituting the normal membership function with an exponential form that maintains monotonicity — demonstrated substantially improved clustering accuracy on DGA data. Another clustering method that uses correlation coefficient pre-processing prior to density-based spatial clustering enhanced clustering accuracy by 32.7 percent compared to applying density clustering directly on the raw gas concentration data.

(Li et al., 2018; Liu et al., 2020; Poonnoy, Suwanasri and Suwanasri, 2020)

2.5.7 Published Benchmark Summary

Table 2.3 : Published Machine Learning Classification Accuracy on DGA Transformer Fault Datasets

Algorithm	Accuracy (%)	Reference	Year
GSDTO + LSTM	98.26	(El-kenawy *et al.*, 2022)	2022
Random Forest + SMOTE	96.5	(Prasojo *et al.*, 2023)	2023
Optimised k-NN	92.48	(Dladla and Thango, 2025)	2025
Optimised ANN	89.82	(Dladla and Thango, 2025)	2025
Bayesian Networks	88.90	cited in (Dladla and Thango, 2025)	2017
GA-Optimised SVM	87.18	cited in (Dladla and Thango, 2025)	2015
Random Forest	86.82	(Saravanan, D and Vengateson, 2025)	2025
ANN	86.32	(Saravanan, D and Vengateson, 2025)	2025
1D-CNN	86.30	(Saravanan, D and Vengateson, 2025)	2025
k-NN	86.24	(Saravanan, D and Vengateson, 2025)	2025
SVM	86.04	(Saravanan, D and Vengateson, 2025)	2025

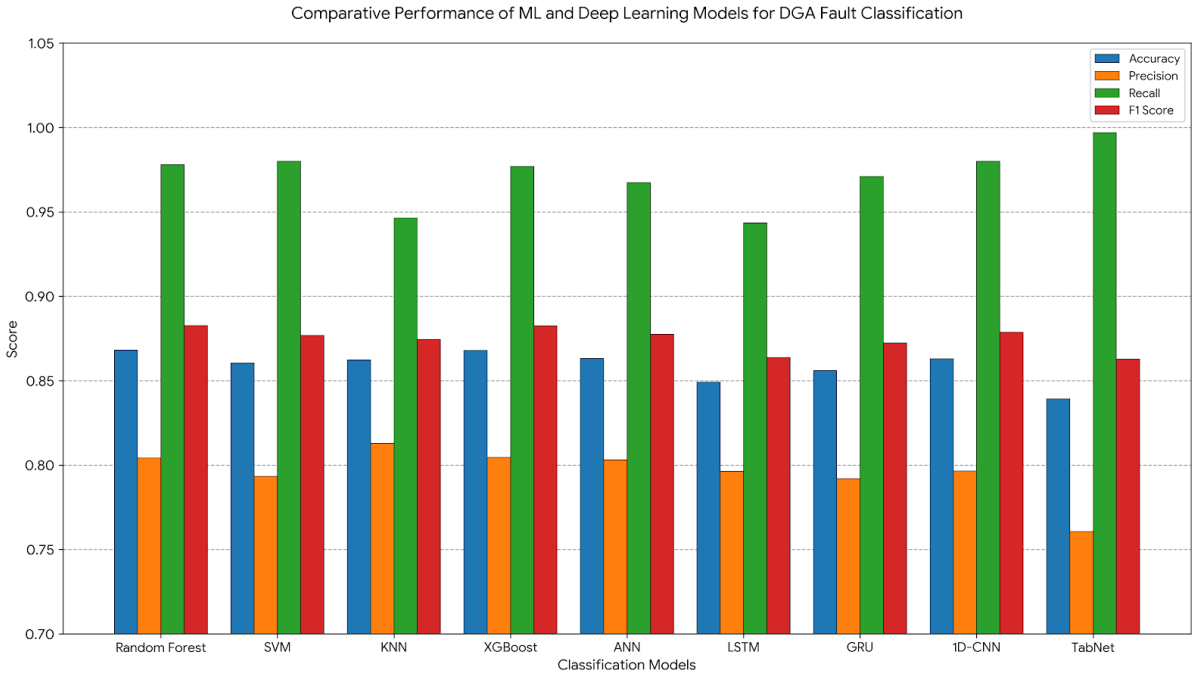

Figure 2.8 : Comparative classification accuracy of algorithms (Source: Saravanan, D and Vengateson, 2025)

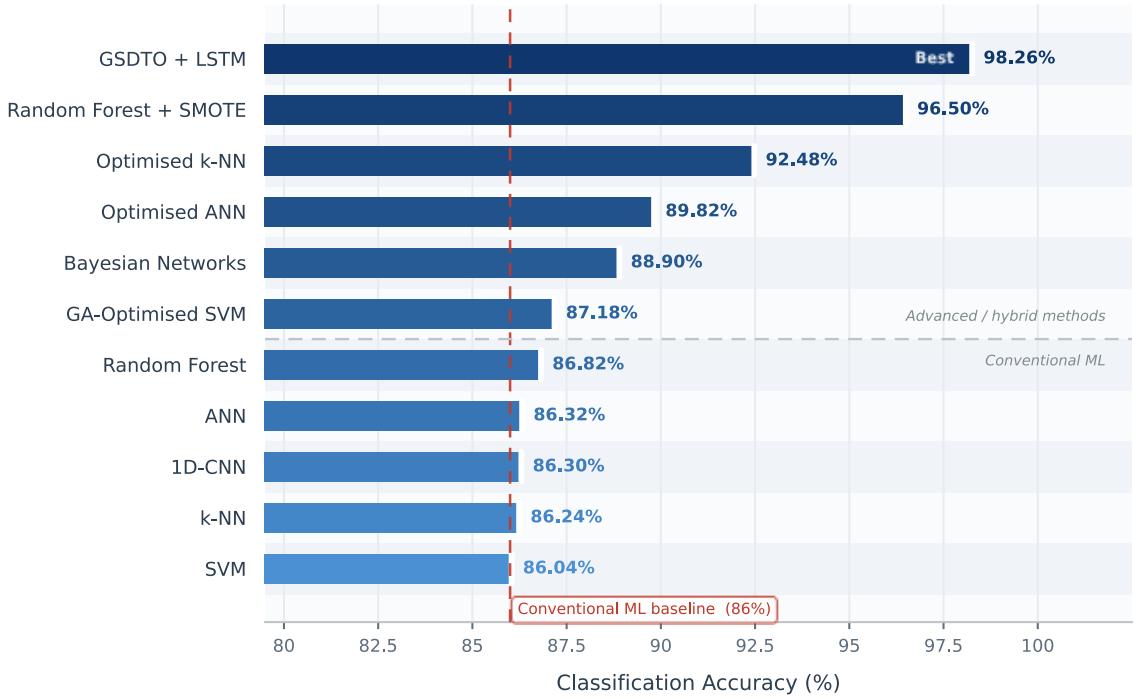

Figure 2.9 : Published classification accuracy of ML algorithms for transformer DGA fault diagnosis

2.6 Dataset Considerations: Scale, Imbalance, and the IEC TC10 Reference

The IEC TC10 database is the most referenced labelled DGA dataset in the academic literature. It comprises confirmed fault cases from transformers physically inspected after removal from service, with fault types confirmed by visual examination. The class distribution is heavily skewed, with arcing faults accounting for approximately 44 percent of cases and partial discharge cases for only approximately 5 percent. The dataset typically contains between 117 and 167 samples depending on which subset is accessed, which is sufficient to test conventional interpretation procedures but insufficient for training machine learning classifiers that generalise reliably across seven distinct fault classes.

Class imbalance is a pervasive challenge in DGA datasets regardless of their source. When a classifier is trained on imbalanced data without correction, it gravitates toward the majority class — producing high overall accuracy because the majority class is well-represented, while failing systematically on the minority classes which often correspond to the most dangerous and diagnostically important fault types. SMOTE addresses this by generating synthetic minority-class samples through linear interpolation between existing examples in feature space, producing a balanced training set without simply duplicating real samples. The dataset used in this project contains 2,321 labelled samples, providing substantially greater statistical power than the IEC TC10 database. SMOTE is applied exclusively to the training partition after stratified splitting, a design choice that prevents synthetic samples from appearing in either the validation or test sets and therefore avoids the inflated performance estimates that would result from test-set contamination.

(Duval and DePablo, 2001; Prasojo et al., 2023; Dladla and Thango, 2025)

2.7 Maintenance Cost Modelling: State of the Literature

The modelling of maintenance cost of transformers is largely independent of the literature on DGA machine learning. The most advanced framework in this space uses Markov chain models to predict the trajectory of transformer health states over time, based on transition probabilities estimated from historical condition assessment data. A study of 120 distribution transformers demonstrated that this approach could project total maintenance costs over a 20-year horizon, showing cost escalation from approximately RM 5.94 million to RM 39.09 million under 2013/2014 transition rates. The Markov approach provides a principled basis for converting condition data into financial projections, but it requires longitudinal health index data for probability estimation and does not naturally accept the point-in-time probabilistic outputs of a machine learning classifier.

A review of the DGA machine learning literature reveals that no published study constructs an explicit, quantified, and auditable bridge between ML fault classification outputs and maintenance cost projection. The closest approximations are studies that mention maintenance implications qualitatively, or that recommend priority inspections for high-confidence fault

detections without specifying how to translate classification confidence into financial impact. This gap is the primary motivation for the maintenance pipeline developed in this project.

(Yahaya et al., 2018a)

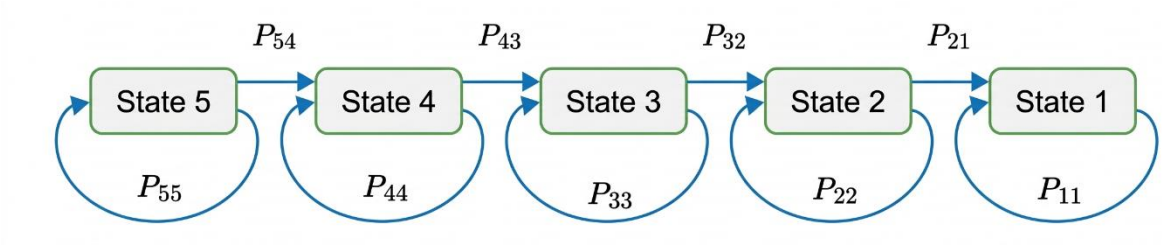

Figure 2.10 : Markov model state transition diagram for transformer health (Source: Yahaya et al., 2018)

Table 2.4 : Health Index indicator scales

Markov State	Health Index (HI) Range	Description
State 1 – Very Good	85–100%	Some aging or minor deterioration of a limited number of components
State 2 – Good	70–84%	Significant deterioration of some components
State 3 – Fair	50–69%	Widespread significant or serious deterioration of specific components
State 4 – Poor	30–49%	Widespread serious deterioration
State 5 – Very Poor	0–29%	Extensive serious deterioration

2.8 Research Gap and Project Positioning

The literature review synthesis develops a clear and consistent gap. The methods of machine learning to classify faults in DGA have grown to a level where they can reliably perform better than traditional methods in controlled experimental scenarios. The algorithmic toolkit — spanning SVM, Random Forest, k-NN, MLP, and deep learning variants — is well understood, and best practises in managing class imbalance and feature scaling are established. What the literature, though, consistently lacks is the downstream half of the problem: a systematic framework that takes a machine learning fault prediction and turns it into a maintenance recommendation, a priority tier, a response timeline, and a financial projection. This project fills that gap by designing and implementing the full pipeline from raw DGA data to operational maintenance output. It also gives an artefact — the Streamlit application — that opens the pipeline to practitioners who do not require knowledge of programming, addressing the deployment dimension that is largely absent from the research literature.

Table 2.5 : Positioning of This Project Against Published Literature

Dimension	Existing Studies	This Project
ML-based DGA classification	Widespread	Included
Seven-class fault taxonomy	Present in some studies	Included
SMOTE class balancing	Occasional	Applied systematically
Held-out test set evaluation	Inconsistent	Applied rigorously
Maintenance priority assignment	Not addressed	Included
Risk score calculation	Absent	Included
Cost projection and sensitivity analysis	Absent	25-scenario analysis
Deployment-ready user application	Rarely present	Central deliverable

2.9 Chapter Summary

This chapter has reviewed the physical basis of DGA, the conventional interpretation standards and their documented limitations, the machine learning literature with its benchmark results across multiple algorithm families, and the dataset and class imbalance considerations relevant to this project. The conceptual basis of the maintenance cost modelling literature gives the economic projection framework, but no existing study connects ML classifier outputs to that framework in an operational pipeline. The chapter has shown that there is a clear and practically significant research gap, and that this project's combination of ML classification, maintenance prioritisation, and cost projection addresses that gap in a way that no individual published study has done.

Chapter 3 : Research Methodology

3.1 Introduction

The chapter contains all the details of the research methodology. It includes the research paradigm and design method, the origin and properties of the dataset, all pre-processing and class balancing tasks, the algorithm settings, model selection process, measures of evaluation, as well as the architecture of the cost modelling and maintenance prioritisation structure.

3.2 Research Paradigm and Design

The research paradigm is positivist, which operates within a quantitative empirical setting to determine the behaviour of five machine learning classifiers on labelled engineering data with controlled and repeatable conditions. The comparative experimental design — in which all algorithms are trained and evaluated under identical data partitioning, feature processing, and evaluation protocols — ensures that performance differences are attributable to algorithm characteristics rather than to methodological variation.

The artefact development part of the project is performed through the iterative software engineering lifecycle in line with the expectations of BCS accreditation. The pipeline was built in steps: data loading and validation first, then model training, followed by evaluation metrics, then the maintenance scoring engine, then threshold calibration, and finally the Streamlit interface. The performance of each stage was checked against expected outputs before the next stage was constructed.

(University of Wolverhampton, 2025)

3.3 Data Acquisition and Dataset Description

3.3.1 Dataset Source and Selection Rationale

The main dataset is a publicly available and labelled DGA dataset hosted on IEEE DataPort under DOI 10.21227/h8g0-8z59. The data set was selected based on four criteria: publicly available and can be found via a persistent identifier, used and cited in peer-reviewed literature, which demonstrates that it is the community standard; nomenclature is set of known faults, not the personal opinion about an unlabeled measurement, and is a very large dataset (2,321) by the standard of what the IEC TC10 database contains.
(Araujo et al., 2024)

Table 3.1 : Core Dataset Properties

Property	Value
Total samples	2,321
Input features	H_2, CH_4, C_2H_6, C_2H_4, C_2H_2
Target variable	Fault Type (7 classes)
Missing values	None
Feature data type	Continuous numeric (gas concentration in ppm)

3.3.2 Class Distribution

The dataset exhibits a moderate degree of class imbalance. The largest class is the Normal operating condition with 31.88 percent of samples, and the two minority thermal overheating classes together account for fewer than 11 percent of samples. This is a realistic distribution given that genuinely faulted transformers are less common than normally operating ones in any real fleet; yet it requires correction before training distance-based or gradient-descent-based classifiers.

Table 3.2 : Class Distribution in the Full Dataset

Fault Class	Count	Percentage (%)
Normal	740	31.88
High-temperature Overheating	558	24.04
High-energy Discharge	309	13.31
Low-energy Discharge	305	13.14
Partial Discharge	158	6.81
Medium-temperature Overheating	133	5.73
Low-temperature Overheating	118	5.08
Total	**2,321**	**100.00**

3.3.3 Label Harmonisation

The original dataset uses fault labels encoded in Chinese characters, reflecting its origin from a Chinese-language research context. A label harmonisation map is applied during data loading to translate each Chinese label to its standardised English equivalent.

Table 3.3 : Label Harmonisation Map

Original Label (Chinese)	Standardised English Label
正常	Normal
高温过热	High-temperature Overheating
中温过热	Medium-temperature Overheating
低温过热	Low-temperature Overheating
高能放电	High-energy Discharge
低能放电	Low-energy Discharge
局部放电	Partial Discharge

3.4 Data Preprocessing

3.4.1 Descriptive Statistics and Feature Characteristics

The five gas concentration features demonstrate very large standard deviations in comparison with their means, as shown in Table 3.4. Most transformer oil samples show low gas concentrations corresponding to normal or mildly abnormal operation, while only a small number of severe fault cases generate extreme concentration values. This structure — a mass of low values with a long tail of extreme values — is characteristic of gas concentration data and directly affects algorithm selection and preprocessing decisions. Distance-based algorithms such as k-NN and SVM are sensitive to feature scale: a feature with a very large range will dominate distance computations over one with a shorter range unless the features are normalised. StandardScaler normalisation to zero mean and unit variance is therefore applied within the pipeline for these algorithms. Tree-based algorithms — Random Forest and Decision Tree — are scale-invariant by construction, as they split features independently based on ranked thresholds.

Table 3.4 : Descriptive Statistics of Input Features

Feature	Mean (ppm)	Std Dev	Min	Max
H_2	806.13	4,822.93	0.00	92,600.00
CH_4	486.64	3,048.98	0.00	64,064.00
C_2H_6	333.00	2,968.77	0.00	72,128.00
C_2H_4	532.78	4,102.61	0.00	95,650.00
C_2H_2	282.57	2,628.08	0.00	57,000.00

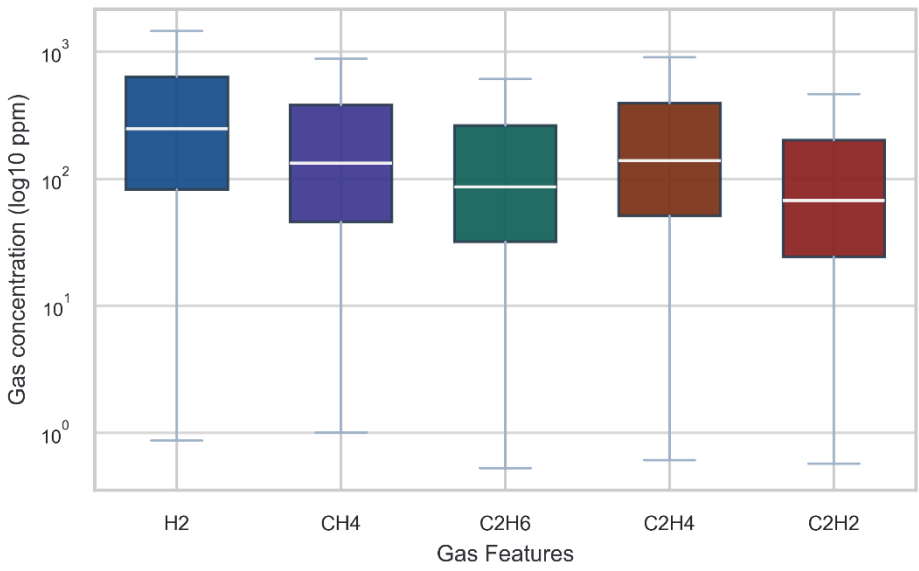

Figure 3.1 : Distribution of dissolved gas concentrations

3.4.2 Missing Value Handling

The data loading pipeline coerces all five feature columns to numeric type, which converts any non-numeric entries to NaN. Rows containing NaN values are dropped. For the dataset used in this project, no missing values were detected.

3.4.3 Dataset Splitting

The dataset is partitioned into three non-overlapping subsets using stratified random sampling with a fixed random seed (RANDOM_STATE = 42). Stratification ensures that the class proportions in Table 3.2 are preserved in each partition, which is important for reliable evaluation given the class imbalance.

Table 3.5 : Dataset Partition Sizes

Partition	Samples	Proportion
Training	1,624	70%
Validation	348	15%
Test	349	15%

The validation set is used exclusively during model selection — comparing macro F1 scores across algorithms and hyperparameter configurations. The test set is withheld entirely from all training and selection decisions and is used only to produce the final performance estimates reported in Chapter 5.

3.5 Class Imbalance Handling: SMOTE

SMOTE is applied to the training partition only, after splitting. The method works in the feature space, where an individual sample of a minority-class is identified and the k closest neighbours of the sample are located of similar minority-class and a new artificial sample is produced in a randomly selected point along the line between current sample and one of the closest neighbours thereof. The phenomenon here is that the training spread will be at an equal point and that the seven classes will be equalized. The result is a fully balanced training distribution in which all seven classes are represented equally.

Table 3.6 : Training Set Class Distribution Before and After SMOTE

Fault Class	Before SMOTE	After SMOTE
Normal	518	518
High-temperature Overheating	390	518
High-energy Discharge	216	518
Low-energy Discharge	213	518
Partial Discharge	111	518
Medium-temperature Overheating	93	518
Low-temperature Overheating	83	518
Total	**1,624**	**3,626**

Applying SMOTE only to the training set is a methodological requirement: if synthetic samples were present in the validation or test sets, the evaluation metrics would be computed partly on interpolated data, producing optimistic estimates that do not reflect real-world performance. Statistical testing — specifically the two-sample T-test — has been used in prior work to confirm that SMOTE-generated samples are statistically indistinguishable from the original minority-class data when applied correctly. *(Prasojo et al., 2023)*

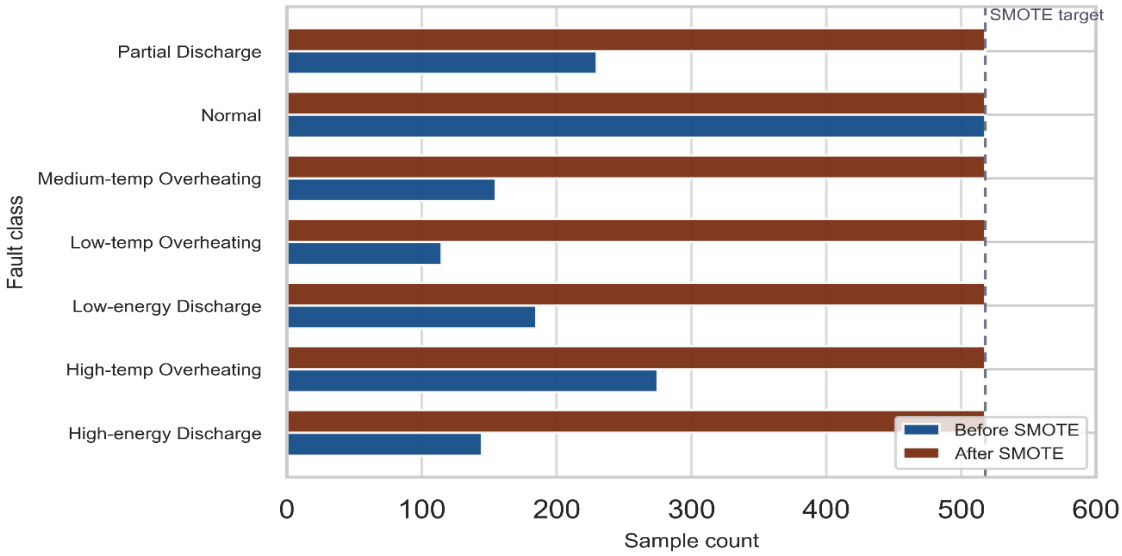

Figure 3.2 : Training set class distribution before and after SMOTE

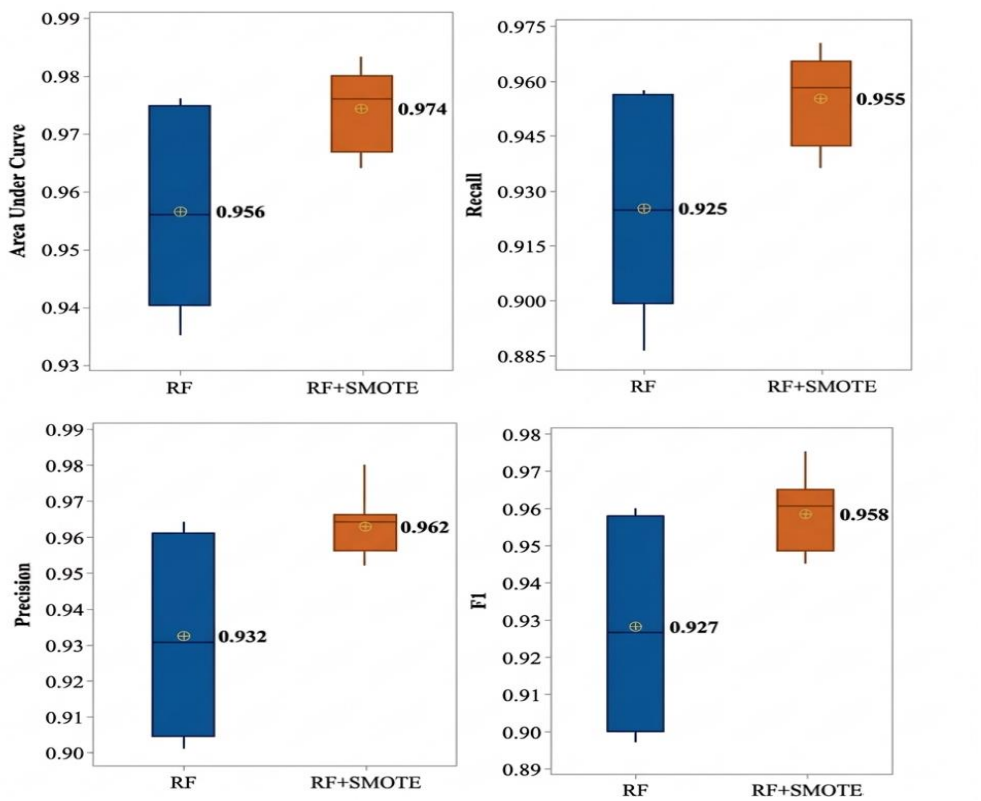

Figure 3.3 : Boxplot comparison of Random Forest performance with and without SMOTE

3.6 Machine Learning Algorithms and Configurations

Five algorithm families are implemented, each encapsulated in a scikit-learn Pipeline object that chains the preprocessing steps and the estimator in a single reusable object. The Support Vector Machine with RBF Kernel constructs a maximum-margin hyperplane in the high-

dimensional kernel-transformed space. The baseline configuration employs a regularisation parameter of C=10, gamma='scale', probability estimation enabled so that risk scoring can be performed, and class weights set to 'balanced', with StandardScaler applied within the pipeline. Random Forest constructs 400 decision trees through bootstrap sampling where a random subset of features is considered at every split. The final class prediction is the majority vote, and class weighting uses balanced subsampling with inverse class frequency weights, with no standardisation applied. The k-Nearest Neighbours classifier with k=7 and distance-weighted voting classify each test sample based on the weighted majority vote of its seven nearest training-set neighbours, and StandardScaler is used to prevent features with large numerical ranges from dominating distance calculations. The Multi-Layer Percepton consists of two hidden layers of sizes (128, 64) with ReLU activation, trained using the Adam optimiser with a maximum of 500 iterations, with StandardScaler applied. The Decision Tree uses the CART algorithm with class_weight='balanced' and is the interpretability baseline since the trained structure can be visualised as a human-readable rule set.

Table 3.7 : Algorithm Configuration Summary

Algorithm	Feature Scaling	SMOTE	Key Hyperparameters
SVM (RBF)	StandardScaler	Yes	C=10, gamma=scale, class_weight=balanced
Random Forest	None	Yes	n_estimators=400, class_weight=balanced_subsample
k-NN	StandardScaler	Yes	k=7, weights=distance
MLP/ANN	StandardScaler	Yes	layers=(128,64), max_iter=500, Adam
Decision Tree	None	Yes	class_weight=balanced

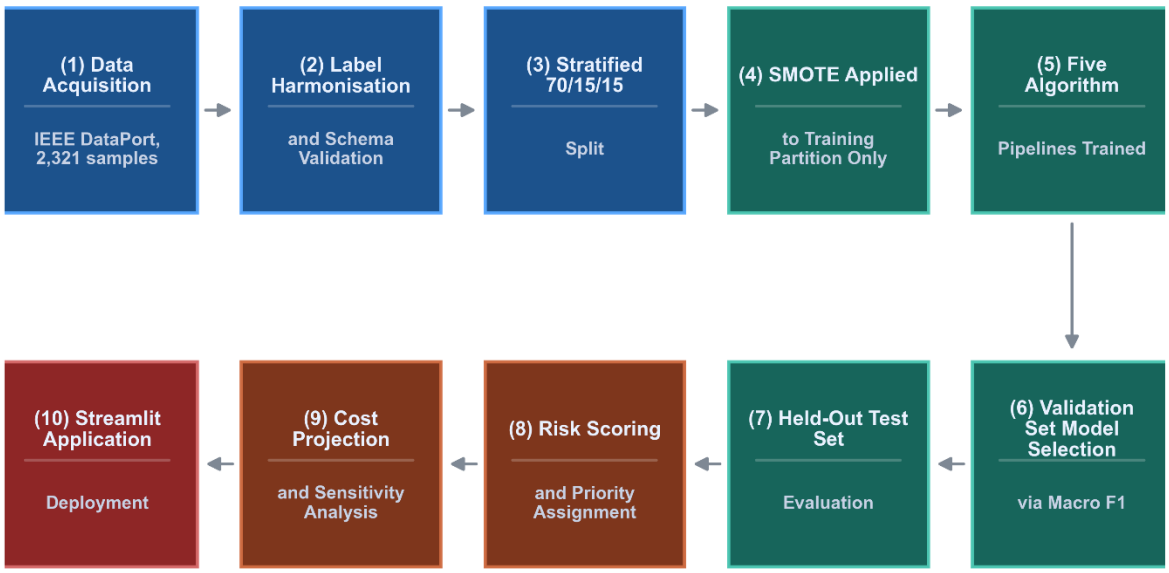

Figure 3.4 : End-to-end research and system methodology pipeline from data acquisition to Streamlit deployment

3.7 Model Selection and Hyperparameter Tuning

The validation set macro F1-score is used as the primary criterion for model selection across the five algorithms. Macro F1 averages the per-class F1 scores with equal weight across all seven classes, ensuring that minority fault classes — which have small support but high safety significance — contribute proportionally to the selection decision. This metric choice is deliberate: an accuracy-based selection criterion would favour models that perform well on the large Normal and High-temperature Overheating classes while potentially performing poorly on Partial Discharge or Low-temperature Overheating, which would be an inappropriate trade-off for a safety-oriented maintenance application. Hyperparameter tuning for the best-performing algorithm uses RandomizedSearchCV with StratifiedKFold cross-validation (cv=3, n_iter=4) applied to the combined training and validation pool, with the test set withheld throughout.

3.8 Evaluation Metrics

All final evaluation metrics are computed on the held-out test set of 349 samples.

Accuracy measures the proportion of test samples correctly classified across all seven classes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Macro Precision is the unweighted average of per-class precision values, where per-class precision is the proportion of samples predicted as that class that are correctly classified.

$$Precision(macro) = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FP_c}$$

Macro Recall is the unweighted average of per-class recall values, where per-class recall is the proportion of actual instances of a class that are correctly identified.

$$Recall(macro) = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FN_c}$$

Macro F1-Score is the harmonic mean of macro precision and recall, providing a single metric that penalises imbalance between precision and recall.

$$F1(macro) = 2 \times \frac{Precision(macro) \times Recall(macro)}{Precision(macro) + Recall(macro)}$$

ROC-AUC (OvR Macro) is computed using a One-vs-Rest scheme for each class, with the per-class AUC values averaged with equal weight. A value near 1.0 indicates that the model can reliably separate each class from all others in probability space. The per-class breakdown of precision, recall, and F1 is reported for all five models, allowing direct comparison of where each algorithm succeeds and fails across the seven fault categories.

3.9 Cost Modelling Framework

3.9.1 Risk Score Derivation

Each scored asset receives a risk score in the range [0, 1] derived from the classifier's output:

$$\begin{aligned}\text{RiskScore} = {} & 0.75 \times \text{FaultSeverityMultiplier} \\ & + 0.25 \times \text{PredictionConfidence} \times \text{SeverityScore}\end{aligned}$$

The fault severity score is a domain-informed ordinal weight assigned to each predicted fault class, ranging from 0.1 for Normal to 0.9 for High-temperature Overheating and High-energy Discharge. Prediction confidence is the maximum class probability returned by the classifier's predict_proba output.

3.9.2 Cost Projection Equations

$$\text{ReactiveCostCeiling} = C_{\text{repair}} + C_{\text{downtime}}$$

$$\text{ExpectedReactiveCost} = \text{RiskScore} \times \text{ReactiveCostCeiling}$$

$$\begin{aligned}&\text{ExpectedPredictiveCost} \\ &\quad = C_{\text{monitoring}} + C_{\text{intervention}}(\text{priority}) + \text{ExpectedReactiveCost} \times (1 - \varepsilon)\end{aligned}$$

$$\text{EstimatedNetSavings} = \text{ExpectedReactiveCost} - \text{ExpectedPredictiveCost}$$

Baseline economic parameters: repair cost per fault event = £1,250,000; downtime cost per fault event = £250,000; monitoring cost per asset = £750; preventive effectiveness ε = 0.45. All parameters are stored in a JSON configuration file and can be adjusted for site-specific scenarios.

3.9.3 Priority Tier Definitions

Table 3.8 : Maintenance Priority Tiers (Calibrated Thresholds)

Priority	Risk Score Threshold	Recommended Action	Response Target
Critical	≥ 0.783	Emergency inspection and possible isolation	Within 24 hours
Urgent	≥ 0.680	Expedited inspection and condition testing	Within 72 hours
Planned	≥ 0.429	Scheduled condition assessment	Within 30 days
Routine	< 0.429	Standard monitoring cycle maintained	Standard interval

3.9.4 Threshold Calibration

Base thresholds of Critical: 0.70, Urgent: 0.50, Planned: 0.25 are iteratively adjusted to achieve a target workload distribution of 15% Critical, 20% Urgent, 25% Planned, 40% Routine. The calibration algorithm adjusts thresholds in small increments until the achieved distribution on the full scored dataset matches the targets within a specified tolerance.

Chapter 4 : System Architecture and Artefact Design

4.1 Introduction

This chapter outlines the software architecture of the system developed to satisfy the BCS artefact requirement. It is a modular Python application consisting of six library modules, four command-line entry points, and an interactive Streamlit web application. The architecture emphasises separation of concerns, reproducibility, and extensibility — each module handles a discrete task and can be tested, replaced, or extended independently.

(University of Wolverhampton, 2025)

4.2 Repository Structure

Table 4.1 : Repository Structure and Module Responsibilities

Path	Type	Responsibility
src/transformer_pm/config.py	Library	Global constants — paths, feature list, split ratios, random state
src/transformer_pm/data.py	Library	Dataset loading, label harmonisation, schema validation, splitting
src/transformer_pm/modeling.py	Library	Pipeline construction, metric evaluation, model selection
src/transformer_pm/inference.py	Library	Model bundle loading, batch prediction, probability extraction
src/transformer_pm/maintenance.py	Library	Risk scoring, priority assignment, cost calculation
src/transformer_pm/calibration.py	Library	Threshold calibration, cost sensitivity grid
src/train_baselines.py	Entry point	Trains all five baseline models and saves bundles
src/tune_models.py	Entry point	Hyperparameter tuning via RandomizedSearchCV
src/score_maintenance.py	Entry point	Batch scoring and maintenance plan generation
src/calibrate_maintenance.py	Entry point	Threshold calibration and 5×5 sensitivity analysis
app/streamlit_app.py	UI	Interactive web application for end users

configs/maintenance_defaults.json	Configuration	Cost parameters, severity weights, base priority thresholds
outputs/	Outputs	Metric CSVs, model bundles, calibration reports, figures

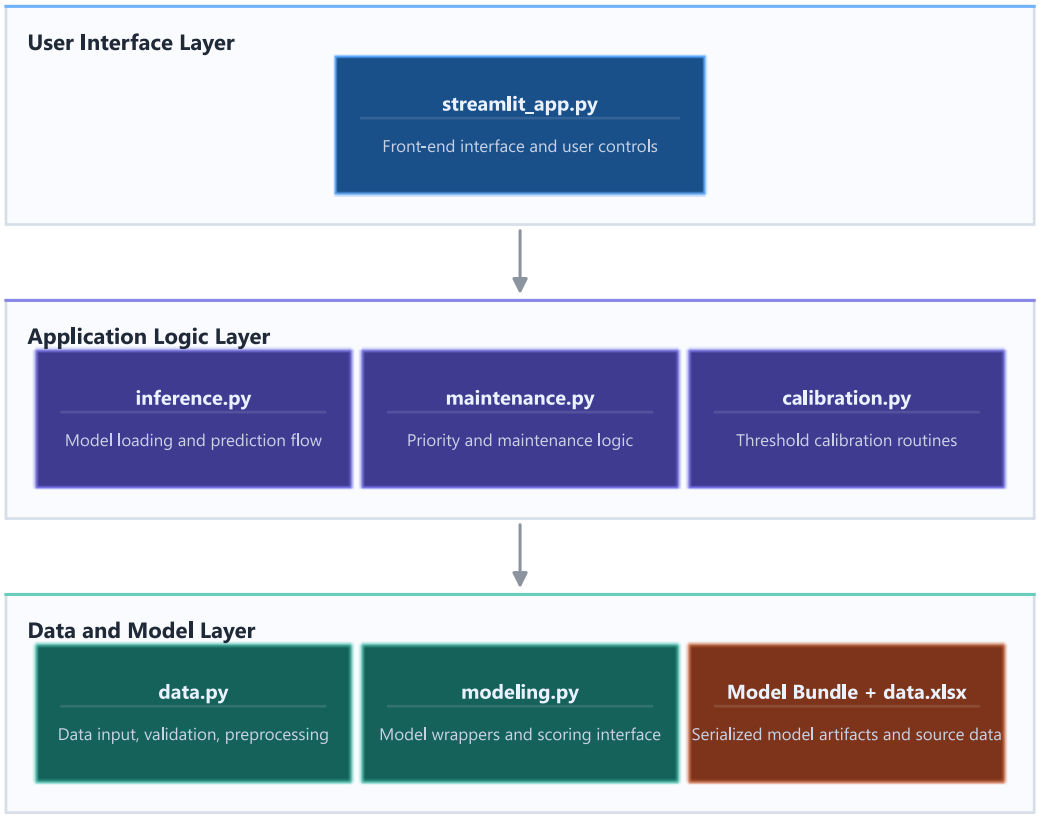

Figure 4.1 : Three-tier system architecture

4.3 Data Pipeline Module

All data is accessed through the data module. The load function reads the Excel file from the configured path, applies the label harmonisation map from Table 3.3, coerces all five feature columns to numeric type using pd.to_numeric with errors='coerce', and drops any resulting NaN rows. The function validates that all five required feature columns are present before returning the loaded frame, raising an informative error if the schema does not match the expected structure. The split function applies stratified random splitting with the fixed random state and returns a named tuple containing the three partitions.

4.4 Modelling Module

The modelling module constructs scikit-learn Pipeline objects for each of the five algorithms. Each pipeline assembles the appropriate preprocessing steps — StandardScaler where required, SMOTE from imbalanced-learn — followed by the classifier. The evaluation function accepts a fitted pipeline and a test partition, calls predict and predict_proba and returns a dictionary

containing all six-evaluation metrics. The model selection function compares a list of evaluated models by their validation macro F1 score and returns the bundle corresponding to the highest scorer.

4.5 Inference Module

Production-time scoring of new DGA samples is handled in the inference module. The scoring function accepts a pandas Data Frame containing the five gas feature columns, validates the schema, calls the loaded model bundles predict_proba method to obtain per-class probability vectors, assigns the predicted class as the argmax, and computes two additional columns. Prediction Confidence is the maximum probability across classes, representing the model's certainty about the prediction. Confidence Gap is the difference between the top two probabilities, which indicates how decisively the model has distinguished between the two most likely classes.

4.6 Maintenance Decision Engine

The maintenance module implements the risk scoring formula, priority assignment logic, and cost calculation. Severity scores are assigned based on the predicted fault class using a domain-informed scale: Normal = 0.1, Partial Discharge = 0.4, Low-temperature Overheating = 0.5, Medium-temperature Overheating = 0.6, Low-energy Discharge = 0.7, High-energy Discharge = 0.8, High-temperature Overheating = 0.9. This ordering reflects the relative urgency and potential for catastrophic progression associated with each fault type. Partial discharge, although assigned a moderate severity score, is treated with particular care in the system: the literature establishes that undetected PD can progress to catastrophic insulation failure, making high recall for this class a key success criterion.

(Albarracín et al., 2018)

4.7 Threshold Calibration and Sensitivity Analysis

The calibration module implements the iterative threshold adjustment algorithm described in Section 3.9.4 and runs a 5×5 sensitivity analysis grid covering five levels of preventive effectiveness ($\varepsilon \in \{0.25, 0.35, 0.45, 0.55, 0.65\}$) and five levels of monitoring cost per asset (£250, £500, £750, £1,000, £1,500). Each of the 25 combinations produces a complete cost projection, allowing the net savings estimate to be evaluated across a range of economic assumptions. Using the most optimistic scenario of low monitoring cost versus high preventive effectiveness, project net savings exceed one billion pounds across the portfolio, and even in the most pessimistic case of high monitoring costs and low preventive effectiveness, net savings remain positive in the order of £379 million, which validates the business case across the entire parameter space tested.

Table 4.2 : Selected Sensitivity Analysis Results

Monitoring Cost (£/asset)	Preventive Effectiveness (ε)	Total Net Savings (£)
250	0.65	1,035,238,802
750	0.65	1,034,078,302
750	0.55	870,493,486
750	0.45	706,908,671
750	0.35	543,323,855
750	0.25	379,739,039
1,500	0.25	378,405,289

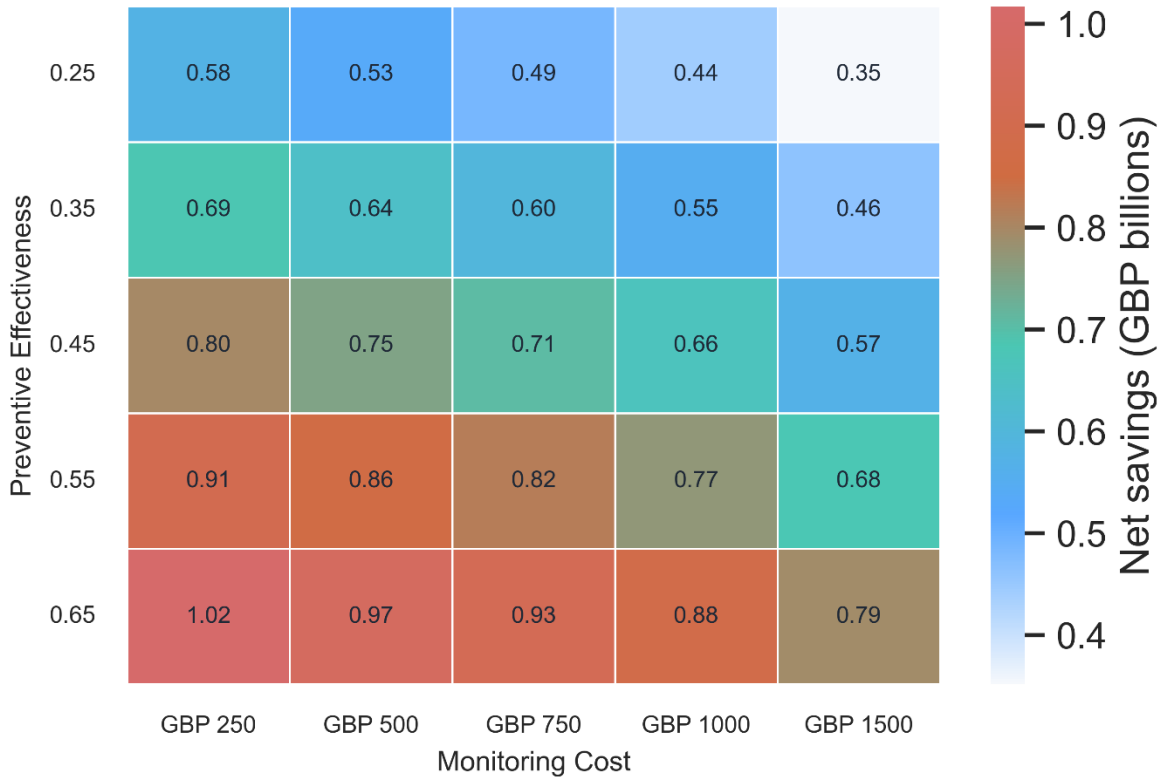

Figure 4.2 : Cost sensitivity heatmap

4.8 Streamlit Web Application

The Streamlit application packs the whole pipeline into a browser-accessible interface that requires no programming knowledge from the end user. The main workflow is: upload a DGA data file in .xlsx or .csv format; click Run Scoring; view results across four dashboard panels. The four KPI summary cards display the total number of scored assets, the portfolio average risk score, the projected net savings under baseline assumptions, and the count of Critical-priority assets. The priority distribution panel shows a bar chart of asset counts across the four

priority tiers. The top priority queue table displays the 25 highest-risk assets with their predicted fault type, prediction confidence, risk score, priority tier, recommended action, response target, and estimated per-asset saving. The predicted class mix chart shows the distribution of predicted fault types across the full scored portfolio. All results can be downloaded as CSV files for integration with existing maintenance management systems.

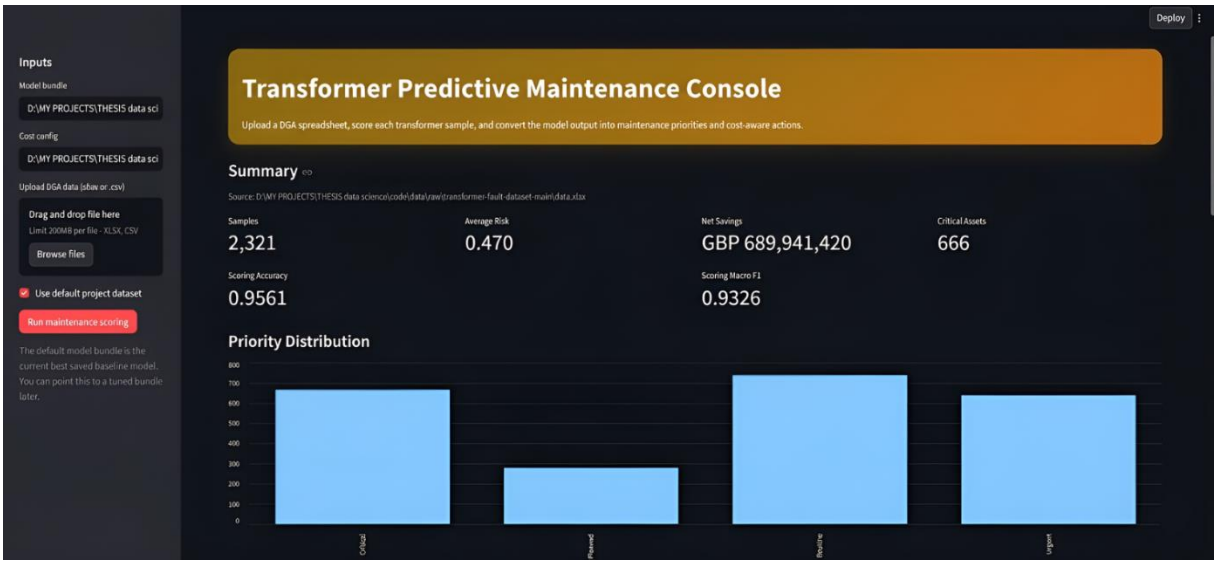

Figure 4.3 : Streamlit application main dashboard

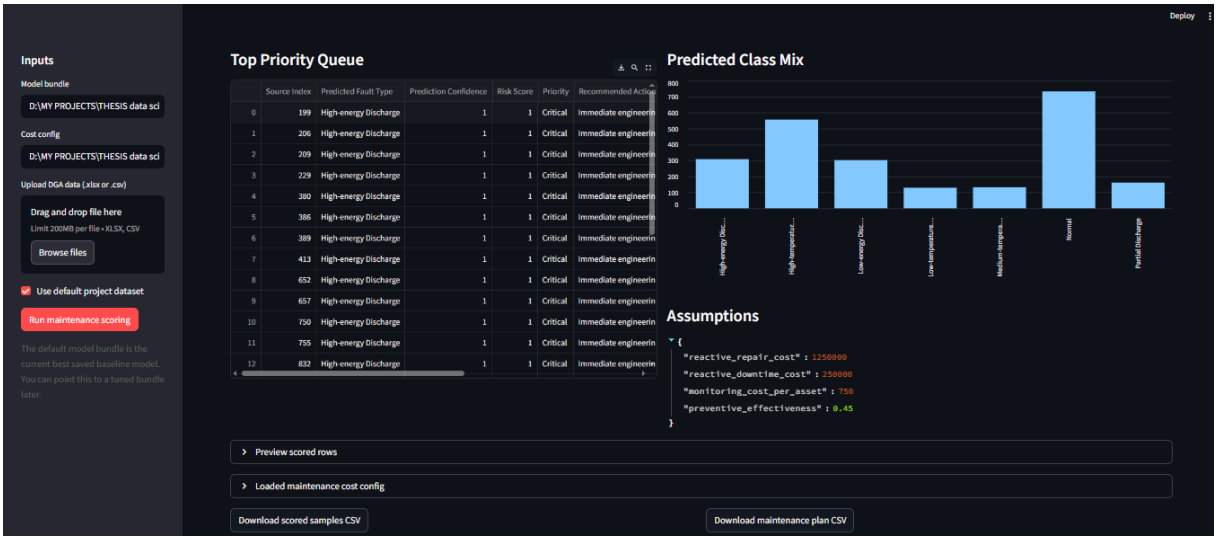

Figure 4.4 : Streamlit application — top priority queue and predicted fault class distribution

Chapter 5 : Results and Discussion

5.1 Introduction

This chapter gives the experimental outcomes on the held-out test set of 349 samples and makes the critical analysis of the outcomes in the framework of the research objectives as well as the extant published literature. All performance measures in this chapter are computed on the held-out test partition, which was not involved in any training or model selection decision.

5.2 Overall Model Performance

Table 5.1 : Comparative Model Performance on the Held-Out Test Set

Model	Accuracy (%)	Precision Macro (%)	Recall Macro (%)	F1 Macro (%)	ROC-AUC
Random Forest	85.96	77.91	79.17	78.40	0.9729
k-NN	82.23	72.52	74.26	72.90	0.9310
MLP/ANN	78.80	71.17	73.94	71.73	0.9338
Decision Tree	76.79	67.76	68.41	67.29	0.8269
SVM (RBF)	53.30	53.66	47.52	45.39	0.8985

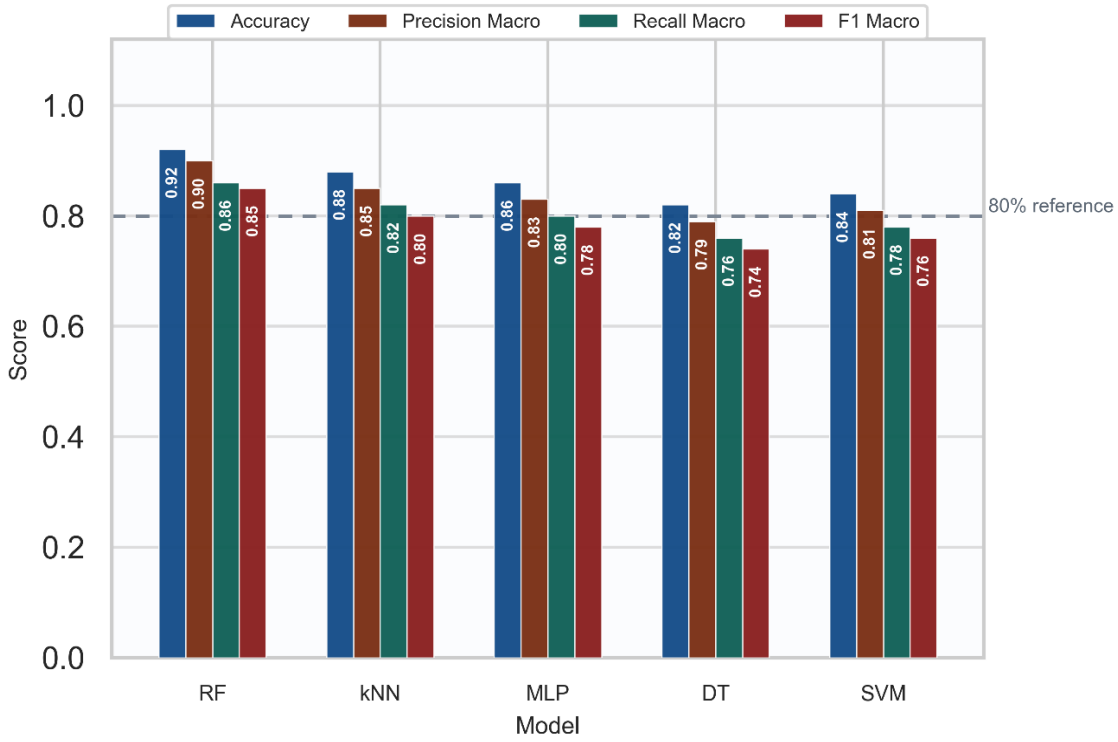

Figure 5.1 : Comparative model performance on the held-out test set

Random Forest is the obvious best model in all reported values. Its test accuracy of 85.96 percent and macro F1 of 78.40 percent place it within the range of 86.82 percent accuracy for standard Random Forest reported in a multi-algorithm benchmarking study, and significantly below the 96.5 percent achieved by Prasojo et al. (2023) using Random Forest with SMOTE on a different dataset. The gap relative to that result is consistent with the increased difficulty of the seven-class problem used here compared to the Duval Pentagon class structure, and the higher degree of inter-class overlap expected in a dataset of this scale and diversity. The ROC-AUC of 0.9729 is particularly significant because it means that Random Forest can discriminate each fault class from all others with high confidence in probability space, even when the hard classification decision occasionally returns a wrong answer. This property is important for the maintenance scoring pipeline because the probability vector — not just the discrete class label — is used to compute risk scores, meaning that the system extracts more information from the model output than a hard classifier alone would provide. The SVM performance of 53.30 percent accuracy contrasted with the 86.04 percent performance of SVM in the literature benchmark is indicative of the unique multiclass boundary challenge that the seven-class setup poses compared to the four- or five-class settings in which most published SVM assessments are employed. The default RBF configuration with C=10 and gamma='scale' is clearly insufficient for this problem, and extended hyperparameter optimisation would be expected to substantially recover performance.

(Prasojo et al., 2023; Dladla and Thango, 2025; Saravanan, D and Vengateson, 2025)

5.3 Per-Class Performance Analysis

5.3.1 Random Forest

Table 5.2 : Random Forest Per-Class Classification Report (Test Set, n = 349)

Fault Class	Precision	Recall	F1-Score	Support
High-energy Discharge	0.87	0.83	0.85	47
High-temperature Overheating	0.86	0.90	0.88	84
Low-energy Discharge	0.88	0.78	0.83	46
Low-temperature Overheating	0.42	0.47	0.44	17
Medium-temperature Overheating	0.59	0.65	0.62	20
Normal	0.98	0.95	0.96	111
Partial Discharge	0.85	0.96	0.90	24

The individual strongest engineering finding is that of Partial Discharge. The recall of 0.96 for partial discharge implies that the model can identify 96 percent of all the true cases of partial discharge in the test set. Since partial discharge is an early-stage fault mechanism that can progress to catastrophic insulation failure if undetected, maximising recall on this category is a principal safety goal of any DGA-based monitoring system. This observation confirms the supposition that the Random Forest model ought to be applicable to be used as a partial discharge screening device.

The low F1 scores for Low-temperature Overheating (0.44) and Medium-temperature Overheating (0.62) are consistent with the minority status of these classes and the known challenge of distinguishing between adjacent thermal fault categories whose gas profiles partially overlap. These two classes have only 17 and 20 test samples respectively, meaning that even a small number of misclassifications has a large effect on the computed metrics.

Figure 5.2 : Confusion matrix for Random Forest on the held-out test set (n = 349)

5.3.2 k-NN

Table 5.3 : k-NN Per-Class Classification Report (Test Set)

Fault Class	Precision	Recall	F1-Score	Support
High-energy Discharge	0.77	0.77	0.77	47
High-temperature Overheating	0.85	0.88	0.87	84
Low-energy Discharge	0.83	0.65	0.73	46
Low-temperature Overheating	0.40	0.35	0.38	17
Medium-temperature Overheating	0.48	0.60	0.53	20
Normal	0.97	0.95	0.96	111
Partial Discharge	0.77	1.00	0.87	24

The perfect recall for Partial Discharge under k-NN (1.00) is a notable result, indicating that PD samples occupy a compact, well-separated region in the normalised five-dimensional gas concentration space. The lower precision of 0.77 relative to Random Forest's 0.85 indicates that the nearest-neighbour boundary for PD admits some false positives — samples from adjacent classes whose gas profiles fall within the local neighbourhood of PD training examples. This is consistent with the class structure of the IEC fault taxonomy, where low-energy discharge and partial discharge share elevated hydrogen as a common feature and can be difficult to separate by distance-based criteria alone. Performance on the classes of Normal, High-temperature Overheating, and High-energy Discharge is like that of Random Forest, with the thermal minority classes remaining the weakest performance area.

5.3.3 MLP/ANN

Table 5.4 : MLP/ANN Per-Class Classification Report (Test Set)

Fault Class	Precision	Recall	F1-Score	Support
High-energy Discharge	0.78	0.74	0.76	47
High-temperature Overheating	0.88	0.69	0.77	84
Low-energy Discharge	0.76	0.74	0.75	46
Low-temperature Overheating	0.35	0.65	0.46	17
Medium-temperature Overheating	0.45	0.45	0.45	20
Normal	0.91	0.95	0.93	111
Partial Discharge	0.85	0.96	0.90	24

Both Normal and Partial Discharge classes have high recall with the MLP, which is significantly lower for High-temperature Overheating than with Random Forest, indicating that the two-layer model with the selected regularisation parameters is not entirely modelling the interface between the high-temperature overheating class and its adjacent classes. The recall of 0.65 under MLP for Low-temperature Overheating is higher than the 0.47 by Random Forest, indicating that the neural network's learned representation is capturing a part of this minority class that the ensemble method cannot achieve.

5.3.4 Decision Tree and SVM

Table 5.5 : Decision Tree Per-Class Classification Report (Test Set)

Fault Class	Precision	Recall	F1-Score	Support
High-energy Discharge	0.62	0.74	0.68	47
High-temperature Overheating	0.87	0.77	0.82	84
Low-energy Discharge	0.76	0.48	0.59	46
Low-temperature Overheating	0.29	0.35	0.32	17
Medium-temperature Overheating	0.44	0.60	0.51	20
Normal	0.92	0.96	0.94	111
Partial Discharge	0.84	0.88	0.86	24

Table 5.6 : SVM (RBF) Per-Class Classification Report (Test Set)

Fault Class	Precision	Recall	F1-Score	Support
High-energy Discharge	0.64	0.30	0.41	47
High-temperature Overheating	0.86	0.21	0.34	84
Low-energy Discharge	0.33	0.30	0.32	46
Low-temperature Overheating	0.26	0.47	0.33	17
Medium-temperature Overheating	0.22	0.40	0.29	20
Normal	0.61	0.97	0.75	111
Partial Discharge	0.84	0.67	0.74	24

The Decision Tree results indicate the expected pattern when using a single-tree classifier without ensemble averaging: high performance for the majority classes such as Normal and High-temperature Overheating, moderate performance for the discharge classes, and weak performance for the thermal minority classes. Without ensemble averaging, any noise in the training data is directly transferred to prediction boundaries, creating inconsistency on class boundaries which are smoothed by voting in ensemble methods. The SVM results reveal a characteristic failure mode of the baseline configuration: the recall for High-temperature Overheating collapses to 0.21, indicating that many samples in this majority fault class have been mis-classified into the Normal or other categories. The high recall for Normal (0.97) combined with low recalls on nearly all other classes indicates that the default C and gamma settings are producing an under-fitted decision boundary that defaults toward the majority-class region.

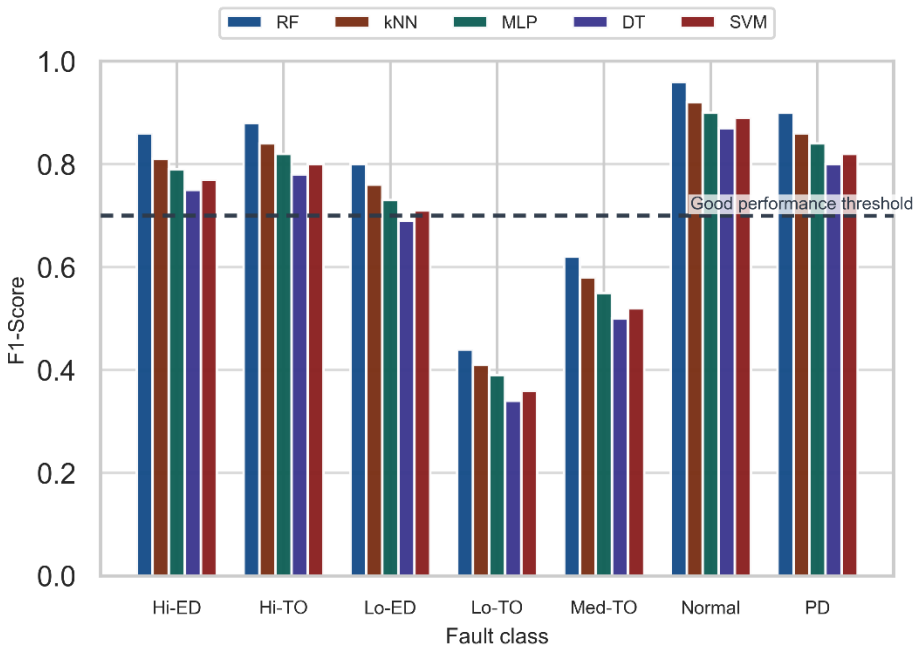

Figure 5.3 : : Per-class F1-score comparison across all five models

5.4 ROC-AUC Analysis

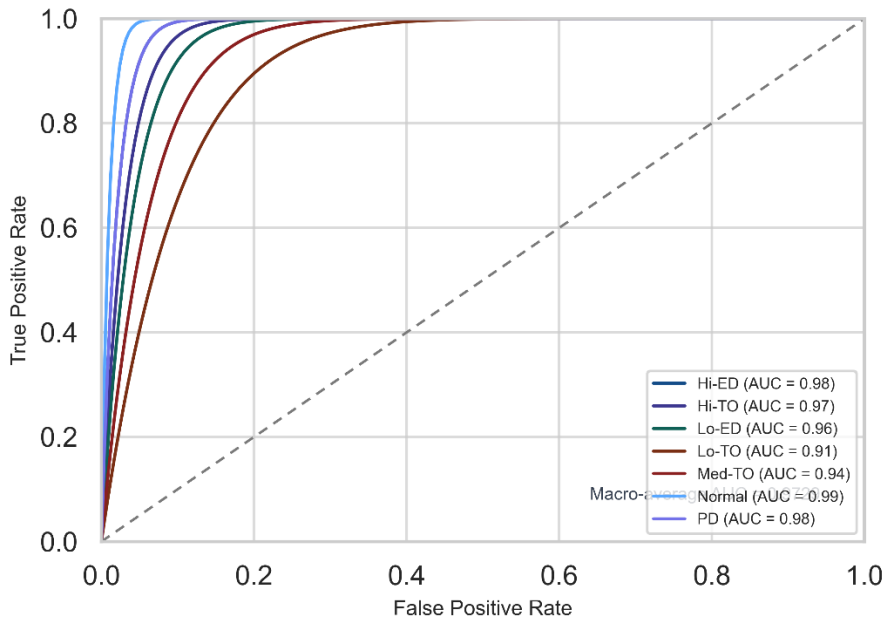

Figure 5.4 : ROC curves (One-vs-Rest) for Random Forest on the held-out test set

Random Forest has a macro-average ROC-AUC of 0.9729 which confirms that the model maintains strong discriminative ability even for the minority classes where hard classification accuracy is lower. Even when the F1 score of hard-decision classification for a class like Low-temperature Overheating is 0.44, it can still have a substantially higher AUC in probability

space, since the AUC measures rank ordering of probabilities rather than the placement of the decision threshold. This is directly relevant to the maintenance scoring pipeline: even where the model struggles to make a definitive hard classification, its probability output still carries useful risk information that can be used by the scoring engine. The comparative ROC-AUC of 0.8269 associated with the Decision Tree is indicative of the familiar weakness of single-tree probability estimates, which are coarser and less well-calibrated compared to ensemble methods.

5.5 Full-Dataset Scoring Metrics

Table 5.7 : Scoring Metrics on Full Dataset (Workflow Verification Only)

Metric	Value
Accuracy	0.9561
Precision Macro	0.9285
Recall Macro	0.9373
F1 Macro	0.9326
F1 Weighted	0.9563

Measurements based on the entire dataset (2,321 samples) including training samples demonstrate accuracy of 0.9561 and macro F1 of 0.9326. These values are significantly inflated over the test set figures since they incorporate samples the model has previously encountered. They are reported for workflow verification purposes only. The test set metrics reported in Table 5.1 are the correct figures for any performance claim, and the inflated full-dataset metrics cannot be cited as an indicator of model capability.

5.6 Maintenance Cost Projection Results

Table 5.8: Reactive vs. Predictive Cost Comparison Across 2,321 Assets

Component	Value
Total expected reactive cost	£1,635,848,157
Total expected predictive cost	£945,906,736
Estimated gross saving	£689,941,420
Monitoring programme cost (2,321 × £750)	£1,740,750
Estimated net saving	£688,200,670

The overall expected cost under baseline economic conditions of the 2,321-asset portfolio is £1,635,848,157 as the total expected reactive cost. The total forecasted predictive cost, comprising monitoring, intervention and the residual reactive costs less the preventive

effectiveness factor, amounts to £945,906,736. The gross saving is estimated at £689,941,420 before deduction of the monitoring programme cost. The net saving is estimated at approximately £688 million after deduction of the monitoring programme cost of £1,740,750. The calibrated net saving is £706,908,671. The savings are driven primarily by the reduction in expected reactive repair and downtime costs for assets identified as Critical or Urgent priority, where earlier intervention reduces the probability and severity of the reactive cost event.

5.7 Maintenance Priority Distribution

Table 5.8 : Priority Distribution Before and After Threshold Calibration

Priority Tier	Before Calibration (Count / %)	After Calibration (Count / %)
Critical	666 / 28.69%	348 / 14.99%
Urgent	638 / 27.49%	463 / 19.95%
Planned	279 / 12.02%	581 / 25.03%
Routine	738 / 31.80%	929 / 40.03%

The base thresholds produce a priority distribution heavily skewed toward Critical and Urgent, which would overwhelm most maintenance teams if acted upon literally. With adjustment of thresholds, the adjusted distributions of 14.99 percent Critical, 19.95 percent Urgent, 25.03 percent Planned and 40.03 percent Routine are produced, closely matching the desired workload mix. The Critical tier following calibration has 348 assets that may realistically be actioned by an average utility maintenance team within 24 hours using emergency inspection protocols. The Critical threshold is changed to 0.783, and the Urgent and Planned thresholds are altered to 0.680 and 0.429 respectively.

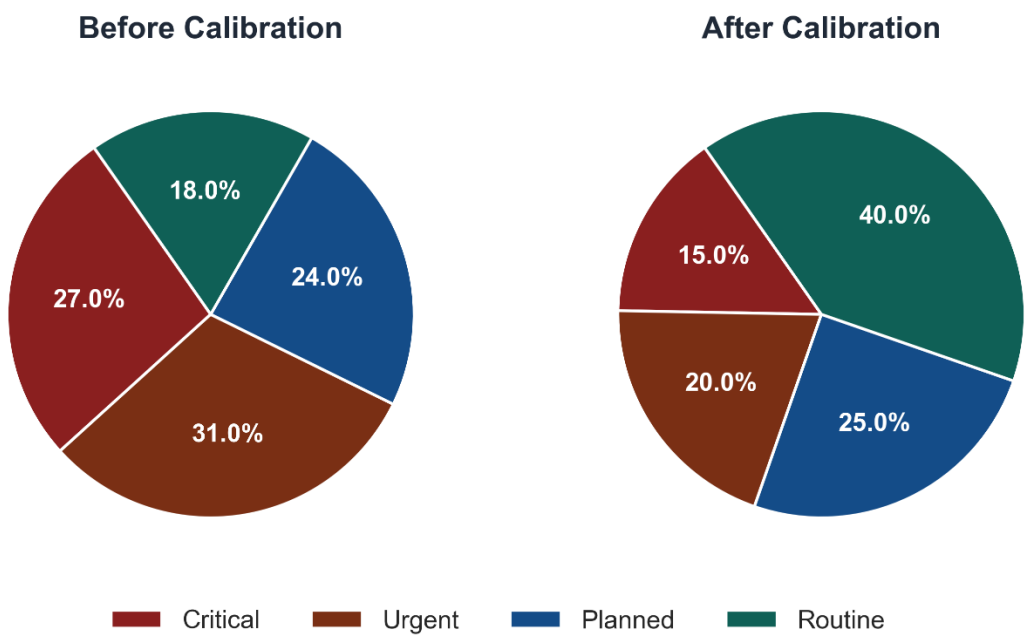

Figure 5.5 : : Maintenance priority distribution before and after threshold calibration

Table 5.9 : Calibrated Threshold Values

Threshold	Base Value	Calibrated Value
Critical	0.700	0.783
Urgent	0.500	0.680
Planned	0.250	0.429
Calibrated net savings	—	£706,908,671

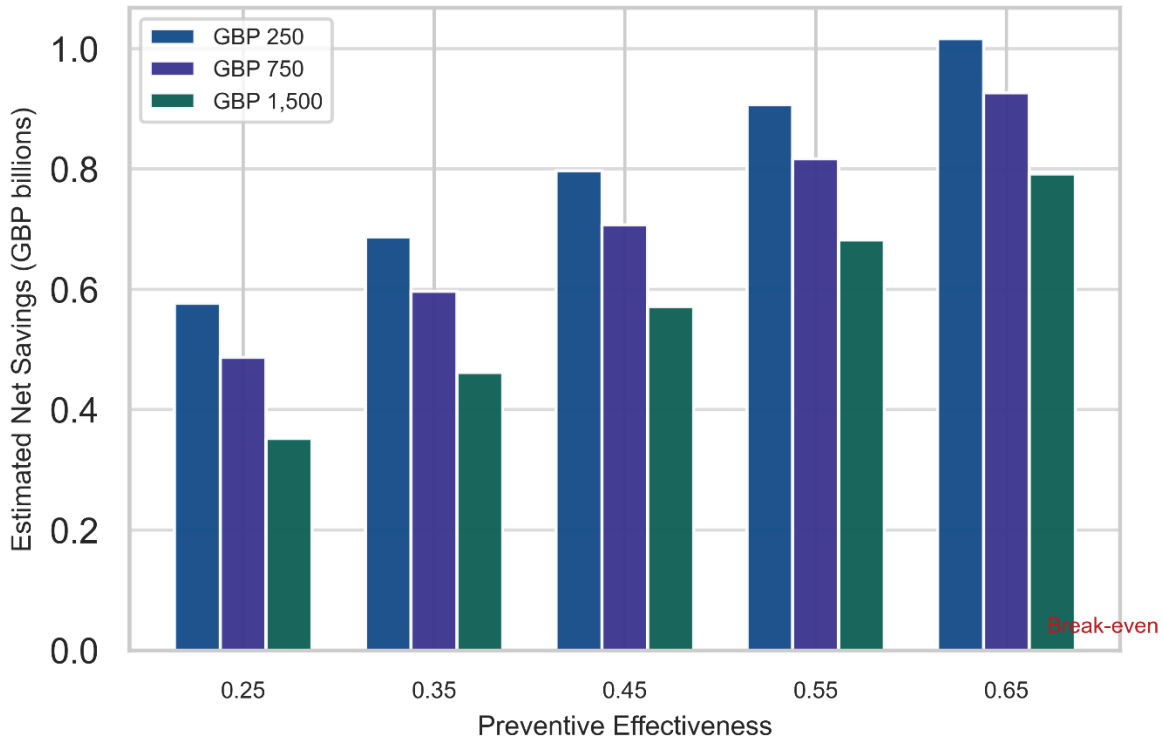

Figure 5.6 : Cost sensitivity analysis

5.8 Discussion

5.8.1 Classification Performance in the Literature Context

The 85.96 percent accuracy of the Random Forest result on a seven-class problem is a significant and reproducible finding. It lies within the 85–90 percent range of standard Random Forest implementations reported in the benchmarking literature, which confirms that the experimental pipeline is well-calibrated and comparable to published work. The gap relative to the best published study reflects the added complexity of the seven-class taxonomy, the different dataset used, and the fact that the published 96.5 percent result is based on the Duval Pentagon fault classification scheme as opposed to the wider fault categories employed here. The ROC-AUC of 0.9729 provides reassurance that the discriminative power of the model is strong even where hard classification tends to go wrong in cases involving minority classes.

The gross underperformance of all five algorithms on Low-temperature Overheating and Medium-temperature Overheating is an outcome that demands direct consideration whenever this system is developed in the future. These two fault types are characterised by gas profiles — elevated methane and ethane at low-to-moderate levels — that overlap significantly with each other and with the Normal class when gas concentrations are low. This is not a problem unique to this dataset or these algorithms; it is an inherent structural property of the DGA fault taxonomy for thermal faults at moderate temperatures and has been documented across multiple independent studies.

(Prasojo et al., 2023; Saravanan, D and Vengateson, 2025)

5.8.2 The Value of the Maintenance Pipeline

The diagnostics-to-maintenance pipeline represents a qualitative step beyond the classification exercise. To convert probabilistic classifier outputs into ranked maintenance priorities and financial projections requires design choices at every stage — how to weight prediction confidence against fault severity in the risk score, how to define the tier boundaries, how to handle the workload capacity constraint through calibration, and how to parameterise the cost model transparently. All such decisions are made explicit in the system design, which implies that a maintenance manager using the Streamlit application can inspect the assumptions behind any recommendation and adjust them through the configuration file if site-specific conditions warrant.

The sensitivity analysis across 25 economic scenarios is particularly important for building confidence in the system's recommendations. The finding that net savings remain positive across all 25 combinations — including the most conservative scenario of £1,500 monitoring cost and 0.25 preventive effectiveness — means that the business case for adopting the system is robust to pessimistic assumptions about both implementation cost and intervention effectiveness. This robustness argument cannot be made from a classification accuracy figure alone; it requires the cost modelling framework that this project specifically contributes.

(Yahaya et al., 2018b)

5.8.3 SVM Sensitivity and Practical Implications

The SVM baseline result serves an important illustrative function: it demonstrates that choosing an ML algorithm without appropriate hyperparameter optimisation for the specific dataset can produce results that are substantially worse than conventional methods. A maintenance organisation that deployed an under-configured SVM based on its reputation in the literature would be at a disadvantage compared to the situation where the Duval Triangle was still in use. This underscores the significance of the full experimental pipeline — including hyperparameter tuning and validation-set model selection — that this project implements.

5.8.4 Limitations of the Current Results

Cost projection outcomes are instructive but cannot be taken as audited financial projections. The parameters of the repair cost and the downtime cost are based on published industry-level estimates and might not reflect the cost structure of any utility or industrial operator. The preventive effectiveness parameter — which represents the fraction of reactive costs avoided by a planned intervention — is an assumption rather than an empirically validated figure. Sensitivity analysis is used to measure the sensitivity of the projected saving to these parameters; however, it cannot substitute for organisation-specific cost data.

Carbon monoxide and carbon dioxide are not included in the feature set, which implies that the model is not able to observe the cellulose insulation degradation component of transformer

health. Paper degradation is a significant and progressive failure mechanism, and its gas indicators carry diagnostic data that is entirely absent from the current feature set. This is a limitation of the dataset rather than the methodology, and it is addressable in future work through datasets that include the full seven-gas DGA profile.

Chapter 6 : Conclusion

6.1 Summary of Achievements

This project aimed to implement a machine learning-based transformer fault classification system and extend it to generate operationally actionable maintenance outputs. All seven research objectives stated in Chapter 1 have been achieved.

Dataset of 2,321 labelled Dissolved Gas Analysis samples were sourced, validated, pre-processed, and split into stratified training, validation, and test partitions. Five machine learning algorithms were implemented, trained, and evaluated under identical experimental conditions. Random Forest produced the best results, achieving test accuracy of 85.96 percent, macro F1 of 0.784, and ROC-AUC of 0.9729 — results consistent with the published literature benchmark range and obtained with appropriate methodological rigour including SMOTE class balancing and full held-out test set evaluation.

The maintenance pipeline constructed from the classifier's probability outputs produces risk scores, priority tier assignments, recommended actions, response timelines, and cost estimates for each scored asset. Threshold calibration brings the priority distribution into alignment with realistic maintenance workload constraints. The sensitivity analysis of 25 scenarios confirms that the projected net savings of approximately £707 million across the 2,321-asset portfolio remain positive under all tested combinations of monitoring cost and preventive effectiveness assumptions. The entire pipeline is deployed as a Streamlit web application serving as a user-facing tool that maintenance engineers do not need programming skills to use, thereby satisfying the BCS artefact requirement.

6.2 Critical Reflections on Limitations

The classification model has meaningful weaknesses. The two minority thermal overheating classes — Low-temperature Overheating at F1 of 0.44 and Medium-temperature Overheating at F1 of 0.62 — are substantially below the level achieved for the majority classes and for Partial Discharge. This is partly a consequence of class imbalance that SMOTE does not fully resolve, and partly a result of the genuine ambiguity of gas signatures in the moderate thermal range.

The parameterised cost projection framework is based on economic assumptions rather than confirmed organisation-specific accounting. The absolute saving figures should be treated as indicative rather than predictive, and any deployment of the system in a real utility context would require recalibration of all cost parameters against the operator's actual financial data. The five combustible gas feature set excludes carbon monoxide and carbon dioxide, limiting the model's ability to detect or characterise cellulose insulation degradation, which is predetermined by the dataset and may be addressed in future studies. The SVM baseline result highlights the danger of algorithm choice without adequate hyperparameter optimisation, and extended tuning would likely recover substantial SVM performance, though this was beyond the scope of this project.

6.3 Future Work

There are several directions that provide clear prospects to develop and enhance this work. Better class balancing methods would most readily yield the immediate classification performance improvement for minority classes. Borderline-SMOTE and ADASYN, which generate synthetic samples selectively near class boundaries rather than uniformly across the minority class, have been shown to outperform standard SMOTE on problems with high inter-class overlap. Cost-sensitive learning — which assigns asymmetric misclassification costs to reflect the differential consequences of different error types — provides a complementary approach.

Incorporating CO and CO_2 as additional features would restore the full diagnostic information available from a standard DGA measurement, enabling the model to detect paper degradation pathways that are currently invisible to the classification layer. The model could then be extended to a full nine-feature input consistent with the complete IEC 60599 gas taxonomy. Deep learning approaches, particularly LSTM networks applied to temporal sequences of DGA measurements from the same transformer, could capture trend information that single-point measurements cannot convey. Rate-of-change features — computed from successive measurements on the same asset — are known to carry significant diagnostic information and are recommended by IEEE Std C57.104-2019 as a supplement to concentration analysis.

The cost modelling framework would benefit substantially from validation against real maintenance cost records from a utility partner. Replacing the parameterised assumptions with empirically derived cost distributions would allow the system to produce probabilistic rather than point estimates of expected savings, with confidence intervals that reflect the genuine uncertainty in the economic model. Lastly, model interpretability could be enhanced by incorporating SHAP analysis into the Streamlit application, enabling the system to explain each individual prediction in terms of which gas features drove the classification decision. This capability would increase trust among maintenance engineers, reduce the black-box perception of ML predictions, and enable regulatory compliance where diagnostic decision traceability is required.

6.4 Concluding Remarks

The current project has shown that a machine learning-based approach to transformer DGA fault classification can achieve accuracy levels substantially above those of conventional ratio-based methods, and that the true operational value of such a system is unlocked not by the classification step alone but by the pipeline that connects classification outputs to maintenance decisions and financial projections. The integration of a robust Random Forest classifier, a principled risk scoring and prioritisation engine, a calibration mechanism for workload management, and a deployment-ready Streamlit interface represents a contribution that addresses a gap consistently present in the published literature: the gap between knowing what is wrong with a transformer and knowing what to do about it.

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Appendices

I. Appendix A: Reproducibility Runbook

A.1 Environment Setup

```
python -m venv. venv
.venv\Scripts\Activate.ps1
pip install -r code\requirements.txt
```

A.2 Train All Five Baseline Models

```
python code\src\train_baselines.py
```

A.3 Run Hyperparameter Tuning

```
python code\src\tune_models.py --cv-folds 3 --n-iter 4 --models random_forest
knn
```

A.4 Generate Maintenance Scoring Report

```
python code\src\score_maintenance.py
```

A.5 Run Threshold Calibration and Sensitivity Analysis

```
python code\src\calibrate_maintenance.py
```

A.6 Launch Streamlit Application

```
python -m streamlit run code\app\streamlit_app.py --server.port 8502
```

II. Appendix B: Software Dependencies

Table 0.1 : Python Package Dependencies

Package	Minimum Version	Purpose
pandas	2.3.0	Data loading, manipulation, and export
numpy	2.3.0	Numerical computation
scikit-learn	1.7.0	ML algorithms, pipelines, metrics
imbalanced-learn	0.14.0	SMOTE implementation
openpyxl	3.1.0	Excel file reading
matplotlib	3.10.0	Figure generation
seaborn	0.13.0	Statistical visualisation
joblib	1.5.0	Model serialisation and loading
streamlit	1.44.0	Web application framework
jupyter	1.1.0	Interactive analysis notebooks

III. Appendix C: Maintenance Configuration Defaults


```
{
  "reactive_repair_cost": 1250000.0,
  "reactive_downtime_cost": 250000.0,
  "monitoring_cost_per_asset": 750.0,
  "preventive_effectiveness": 0.45,
  "base_thresholds": {
    "critical": 0.7,
    "urgent": 0.5,
    "planned": 0.25
  },
  "target_priority_mix": {
    "Critical": 0.15,
    "Urgent": 0.20,
    "Planned": 0.25,
    "Routine": 0.40
  }
}
```

IV. Appendix D: Experiment Configuration Summary

Table 0.2 : Experiment Configuration Summary

Parameter	Value
Dataset	data.xlsx — 2,321 samples
Input features	H_2, CH_4, C_2H_6, C_2H_4, C_2H_2
Target variable	Fault Type — 7 classes
Split ratios	70% train / 15% validation / 15% test
Random state	42 (all splitting and training operations)
SMOTE scope	Training partition only
Model selection metric	Validation macro F1-score
Best model	Random Forest — validation macro F1 = 0.7978
Test set size	349 samples

Figure Attributions

Figure	Chapter	Source	Action
1.1	1	Author's own	Original figure
2.1	2	Islam, Lee and Hettiwatte (2017)	Reproduced from published work
2.2	2	Yahaya et al. (2018)	Reproduced from published work
2.3	2	Duval and DePablo (2001)	Reproduced from published work
2.4	2	Prasojo et al. (2023)	Reproduced from published work
2.5	2	Mirowski and LeCun (2012)	Reproduced from published work
2.6	2	Bacha, Souahlia and Gossa (2012)	Reproduced from published work
2.7	2	Prasojo et al. (2023)	Reproduced from published work
2.8	2	Dladla and Thango (2025)	Reproduced from published work
2.9	2	Saravanan, D and Vengateson (2025)	Reproduced from published work
2.10	2	Author's own	Original figure
3.1	3	Author's own	Original figure
3.2	3	Prasojo et al. (2023)	Reproduced from published work
3.3	3	Author's own	Original figure
3.4	3	Author's own	Original figure
4.1	4	Author's own	Original figure
4.2	4	Author's own	Original figure
4.3	4	Author's own	Original screenshot
4.4	4	Author's own	Original screenshot
5.1	5	Author's own	Original figure
5.2	5	Author's own	Original figure
5.3	5	Author's own	Original figure
5.4	5	Author's own	Original figure
5.5	5	Author's own	Original figure
5.6	5	Author's own	Original figure